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Level Process Trainer

ACE 101

Advanced Control Equipments,
Sangli, Maharashtra



User Manual

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1. Product Details

1.1 Description

The Level Process Trainer (ACE 101) is a compact experimental setup designed to help users understand the principles of level measurement and control in process systems. It provides a practical platform for studying how the liquid level in a tank can be measured, monitored, and controlled using standard industrial instrumentation. The system consists of a process tank fitted with a level transmitter that continuously measures the liquid level. Drain valve arrangements are provided to regulate and discharge the liquid when required. The inflow to the tank is controlled by a pneumatically operated control valve that functions on a standard 3–15 psi pressure signal.

As the controller generates a 4–20 mA electrical output, a current-to-pressure (I/P) converter is used to convert this signal into the corresponding pneumatic pressure required to operate the control valve. The liquid level, acting as the process variable, is monitored and regulated using a digital indicating controller with adjustable set points and control parameters.

The entire setup is mounted on a support frame designed for convenient tabletop operation and easy access. The controller can also be connected to a computer through a USB interface for monitoring and control. User-friendly software is provided to conduct experiments and observe process behavior in real time.



Note the following parameters related to the flow process trainer.

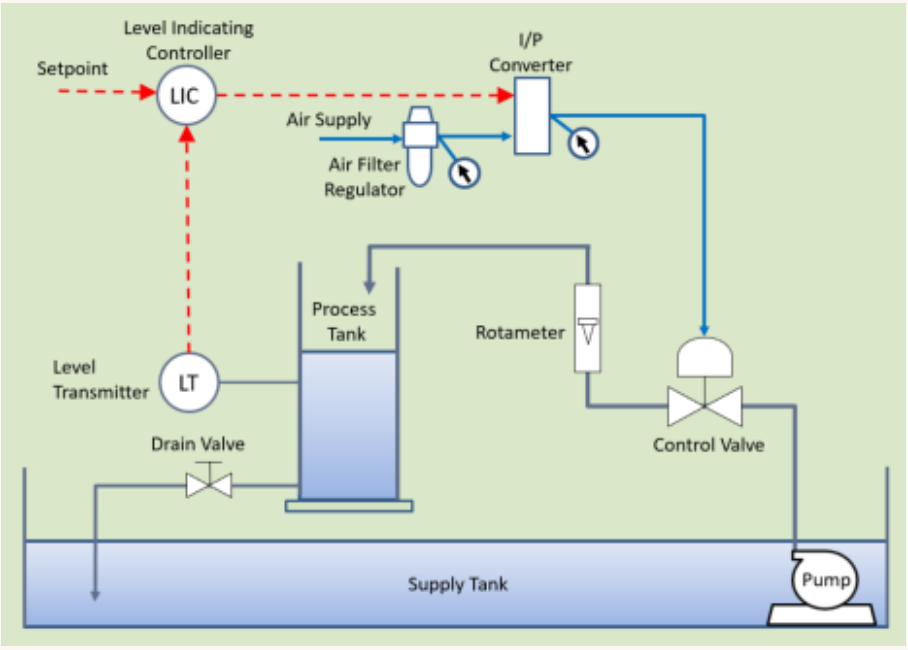
Process Variable (PV)	: Tank Level
Manipulated Variable (MV)	: Inlet Flow (q_i)
Controlled Variable (CV)	: Tank Level
Action	: Direct

1.2. Specification Details

Product	Level Process Trainer
Product code	ACE 101
Type of control	Supervisory Control and Data Acquisition (SCADA)
Control unit	Delta DTB9696 - Digital indicating controller with RS485 (Modbus), Power supply 230 VAC – 1 No.
Supply Tank	SS 304, Size: 150 x 150 x 450 mm with drain valve – 1 No.
Level Transmitter	Pressure Transmitter - Make: Wika, Model: A-10, Output: 4-20 MA, Conn:1/2" BSP M, Accuracy: +/-0.5% BFSL, Temp Range: 0 To 80 Deg. C, Range: 0 to 0.05 Bar – 1 No.
Pump	Submersible, Model: HQB 4500, 400LPH – 1 No.
Control Valve	Linear, Globe 2way, Size 1/2", Screwed Female End, Body CCS, Trim SS, Travel 14.3, CV=0.63, Air to CLOSE, Spring range 0.2-1, Actuator 12 Square inch – 1 No.
Rotameter	Make: Eureka, Model: MG11, Range: 10-100 LPH, Conn.1/2", Bottom-Top, PTFE+Neoprene, Screwed – 1 No.
I/P Converter	Make Control Air Inc, Type T500-AC, input 4-20 mA DC, output 3-15 psig, End connection 1/4 BSP – 1 No.
Airline integral filter regulator	Miniature series, Make: Airmatic, Model MB10-02-1-PAP-PD (Alu body, Polycarbonate bowl, G1/4 BSP, Range 0-2 Kg/cm^2) – 1 No.
Pressure Gauge	2.5 kg/cm2 - 1 No, 7 kg/cm2 - 1 No
Panel	MS powder coated with necessary switch and indicators
Support Structure	Anodized Aluminum Extrusion Rail Frame

1.3 P&I Diagram

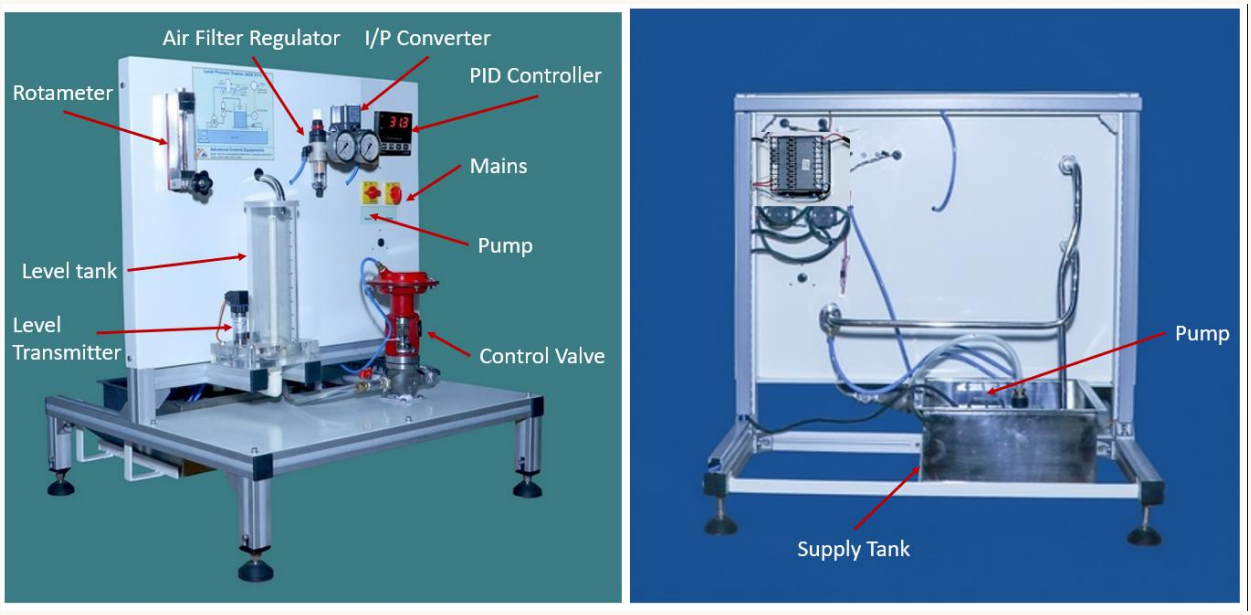
The piping and instrumentation diagram of the Level Process Trainer is shown below:



1.4 Component Details

The front and back view of the trainer with component description is illustrated below

Front & Back View



1.5 Requirements

- Electric supply:
  - 230 VAC single phase electric supply with proper earthing. (Neutral – Earth voltage less than 3 VAC)
  - 5A, three pin socket with switch (1 No.)
- Computer:
  - Standard configuration - Intel Core i5, 8GB RAM, 512GB SSD, Wi-Fi, USB 3.0.
- Operating System
  - Windows 10 / Windows 11 Professional with MS excel (any version) to export data in excel format
- Table Space
  - Size: 400W x 700D x 400H in mm

1.6 Warranty

The setup is covered by a one-year warranty against manufacturing defects from the date of commissioning. Any repairs required during the warranty period will be carried out free of cost. This warranty, however, does not cover damage resulting from mishandling, improper power supply, operation without water, or similar misuse.

2. Process Control

2.1 Essentials of Process Control

Process controls is a mixture between the statistics and engineering discipline that deals with the mechanism, architectures, and algorithms for controlling a process. Some examples of controlled processes are:

1. Controlling the temperature of a water stream by controlling the amount of steam added to the shell of a heat exchanger.
2. Operating a jacketed reactor isothermally by controlling the mixture of cold water and steam that flows through the jacket of a jacketed reactor.
3. Maintaining a set ratio of reactants to be added to a reactor by controlling their flow rates.
4. Controlling the height of fluid in a tank to ensure that it does not overflow.

The Objectives of Control:

- Maintain the process at the operational conditions and set points
- Transition the process from one operational condition to another

Definitions:

Input Variable - This variable shows the effect of the surroundings on the process. It normally refers to those factors that influence the process. An example of this would be the flow rate of the steam through a heat exchanger that would change the amount of energy put into the process. There are effects of the surrounding that are controllable and some that are not. These are broken down into two types of inputs.

a. Manipulated inputs: variable in the surroundings can be control by an operator or the control system in place.

b. Disturbances: inputs that cannot be controlled by an operator or control system. There exist both measurable and immeasurable disturbances.

Output variable- Also known as the control variable These are the variables that are process outputs that effect the surroundings. An example of this would be the amount of CO_2 gas that comes out of a combustion reaction. These variables may or may not be measured.

As we consider a controls problem. We are able to look at two major control structures.

- Single input-Single Output (SISO) - for one control (output) variable there exist one manipulate (input) variable that is used to affect the process.
- Multiple input-multiple output (MIMO) - There are several control (output) variable that are affected by several manipulated (input) variables used in a given process.

Terminology:

The terms commonly used in process control is listed out and a brief description is given below.

- Bias: It's the amount of controller output when error is zero in a proportional control mode.
- Cascade: A control system with 2 or more controllers, a "Master" and "Slave" loop. The output of the "Master" controller is the setpoint for the "Slave" controller.
- Dead Time: The amount of time it takes for a process to start changing after a disturbance in the system.
- Derivative Control: The "D" part of a PID controller. With derivative action the controller output is proportional to the rate of change of the process variable or error.
- Error: In process controls, error is defined as: Error = setpoint - process variable.
- Integral Control: The "I" part of a PID controller. With integral action the controller output is proportional to the amount and duration of the error signal.
- PID Controller: PID controllers are designed to eliminate the need for continuous operator attention. They are used to automatically adjust system variables to hold a process variable at a setpoint. Error is defined above as the difference between setpoint and process variable.
- Proportional Control: The "P" part of a PID controller. With proportional action the controller output is proportional to the amount of the error signal.
- Setpoint: The setpoint is where you would like a controlled process variable to be.

Practical Considerations:

The following concepts will often be used in practical process control loop / PID controller design.

- Action: It determines the controller output will vary from 'low to high' or 'high to low' for a particular error. It can be explained as, if PV is increasing above setpoint ('high'), controller output (CO) is also increasing ('high'), then it is 'Direct' action. Otherwise if PV is 'low', controller output is 'high', it is 'Reverse' action. Action of the controller also determines whether the controller output will be 0% or 100% during startup from zero condition.
- Auto / Manual Mode: In Auto mode, the controller (software / hardware) will execute the control algorithm and gives the output. Whereas, in manual mode, the control algorithm will not be executed and the controller takes user input as controller output. In a practical scenario, manual mode will be used during startup and once the process variable reaches close to the setpoint, the controller will be shifted from Manual to Auto mode.
- Bumpless Transfer: Bumpless transfer allows the controller to be switched between manual and automatic modes and back again without the control output suddenly 'bumping' to a different value. Bumpless transfer is often implemented by making the setpoint track the PV when in manual mode. On changing

back to auto, the SP and PV will initially be the same (i.e. error=0) so the output will also stay the same. The disadvantage is that the operator has to re-adjust the setpoint back to where it was. There are other ways of doing it. For example, when in manual mode, any change to the control output is added / subtracted from the integral term. When the controller is put back into auto, it starts off with the integrator maintaining the current control output rather than jumping back to the previous value.

- Anti-Reset Windup: Also known as integrator windup or reset windup, refers to the situation in a PID feedback controller where a large change in setpoint occurs (say a positive change) and the integral terms accumulates a significant error during the rise (windup), thus overshooting and continuing to increase as this accumulated error is unwound (offset by errors in the other direction). The specific problem is the excess overshooting.

  This problem can be addressed by:

  - Initializing the controller integral to a desired value, for instance to the value before the problem
  - Increasing the setpoint in a suitable ramp
  - Disabling the integral function until the to-be-controlled process variable (PV) has entered the controllable region
  - Preventing the integral term from accumulating above or below pre-determined bounds
  - Back-calculating the integral term to constrain the process output within feasible bounds.
  - Servo and Regulatory Responses: The effect of Process Variable (PV) for a setpoint change is called as servo response whereas the effect for a load disturbance is called as regulatory response.

References:

- Chemical Process Dynamics and Controls open textbook, University of Michigan.
- http://control.com/thread/1026220163
- https://en.wikipedia.org/wiki/Integral_windup

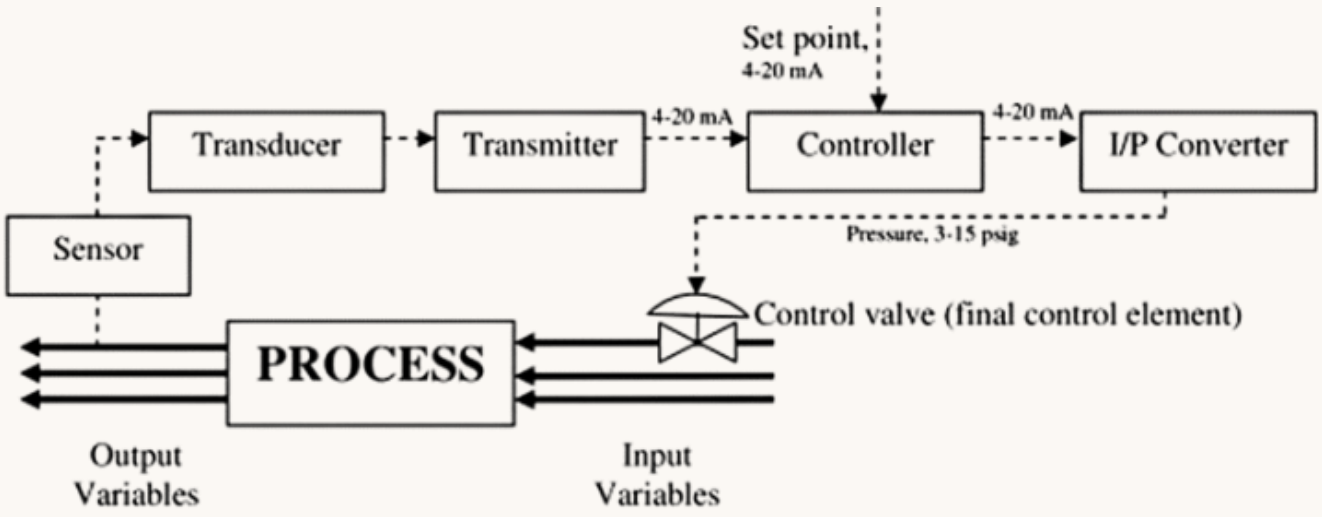
2.2 Feedback Control

Introduction:

The safe and stable operation of a process requires that key variables be maintained at or close to their design values in the face of disturbances entering the process. Maintaining a process variable at or near a certain value requires a manipulation handle that can be appropriately adjusted. This leads to the idea of feedback control where the deviation in the variable to be maintained at / near its design value is used to make appropriate adjustments in the manipulation handle. The variable to be maintained at its design value is referred to as the *controlled variable* and the adjustment handle is called the *manipulated variable*. The algorithm / procedure used to quantitatively translate the deviation in the controlled variable to the adjustment in the manipulated variable is known as the *control algorithm*.

Feedback Control Loop:

A feedback control loop is schematically illustrated in Figure below. Its primary components are the sensor, transducer, transmitter, controller, I/P converter and the final control element. The sensor is the sensing element used to measure the controlled variable (and other important process variables that may not be controlled). Flow, temperature and pressure sensors are routinely used in the process industry.



For standardization across different manufacturers, the range of the input and output signal from a controller is 4-20 mA. The range corresponds to the sensor / final control element span. The transmitter converts the electrical signal from the transducer to the 4-20 mA range. The transmitter signal is input to the controller. The desired value for the controlled variable, referred to as the set-point, is also input to the controller. The controller output signal is again between 4-20 mA.

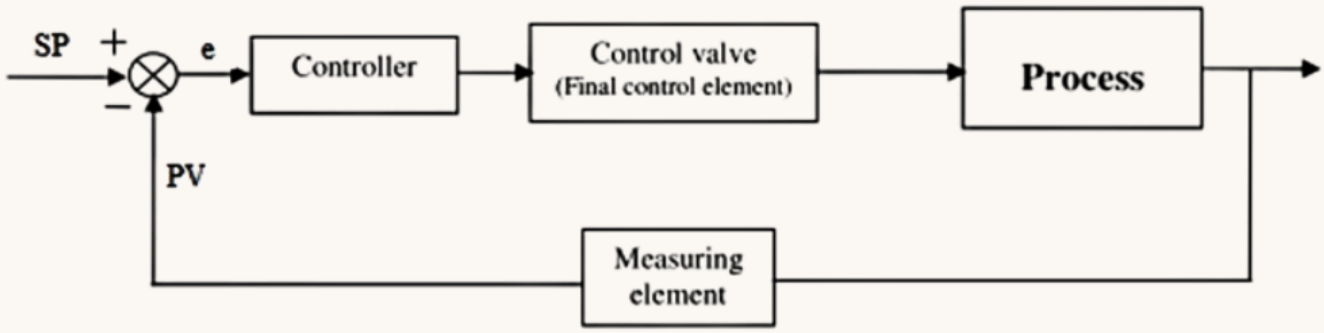
In the process industry, this electrical signal is converted to an equivalent 3-15 psig pneumatic pressure signal using an I/P converter. The pressure signal (or rather change in the pressure signal) is used to move

the final control element to bring about a change in the manipulated variable. In the process industry, almost all final control elements are control valves that adjust the flow rate of a material stream.

The controller subtracts the current value of the controlled variable from its set-point to obtain the error signal as

$$e(t) = SP - PV$$

where PV is the controlled variable. The error signal is input to the control algorithm to determine the change in the manipulated variable (control input) to be implemented. This is schematically illustrated in Figure below.



References:

- http://nptel.ac.in/courses/103104049/2

2.3 Controller Modes

Introduction:

A controller generates a control signal to the final control element, based on a measured deviation of the controlled variable from the setpoint. The choice of operating mode for any given process-control system is a complicated decision. It involves not only process characteristics but cost analysis, product rate, and other industrial factors. At the outset, the process-control technologist should have a good understanding of the operational mechanism of each mode and its advantages and disadvantages.

Two-Position Mode:

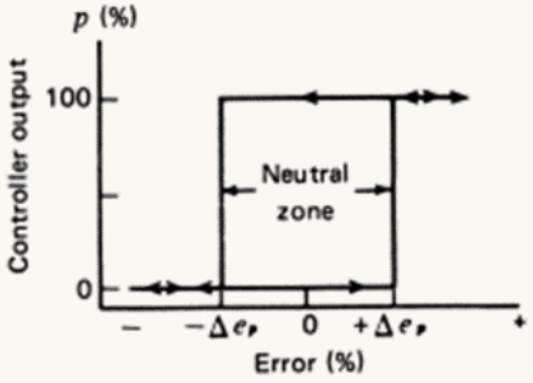
The most elementary controller mode is the ON/OFF, or two-position, mode. This is an example of a discontinuous mode. It is the simplest and the cheapest, and often suffices when its disadvantages are tolerable. Although an analytic equation cannot be written, we can, in general, write as given below.

$$p = \begin{cases} 0\% & e_p < 0 \\ 100\% & e_p > 0 \end{cases}$$

This relation shows that when the measured value is less than the setpoint, full controller output results. When it is more than the setpoint, the controller output is zero. A space heater is a common example. If the temperature drops below a setpoint, the heater is turned ON. If the temperature rises above the setpoint, it turns OFF.

Neutral Zone:

In virtually any practical implementation of the two-position controller, there is an overlap as e_p increases through zero or decreases through zero. In this span, no change in controller output occurs. This is best shown in Figure below, which plots controller output (%) versus e_p for a two-position controller.



We see that until an increasing error changes by Δe_p above zero, the controller output will not change state.

In decreasing, it must fall Δe_p below zero before the controller changes to the 0% rating. The range2 Δe_p, which is referred to as the neutral zone or differential gap, is often purposely designed above a certain minimum quantity to prevent excessive cycling. The existence of such a neutral zone is an example of desirable hysteresis in a system.

Proportional Control Mode:

In this mode, over some range of errors about the setpoint, each value of error has a unique value of controller output in one-to-one correspondence. The range of error to cover the 0% to 100% controller output is called the proportional band, because the one-to-one correspondence exists only for errors in this range. This mode can be expressed by

$$p = K_P e_p + p_0$$

where, P_0 = controller output with no error (%)

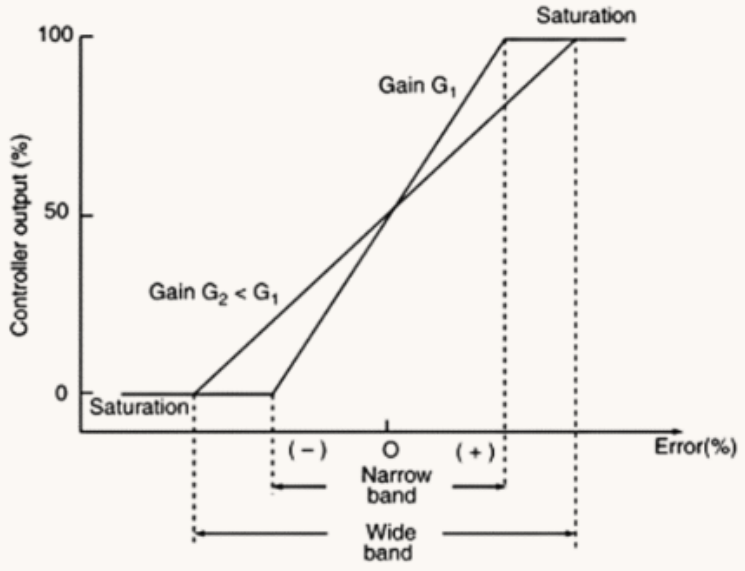
K_p = proportional gain between error and controller output (% per %)

Proportional Band:

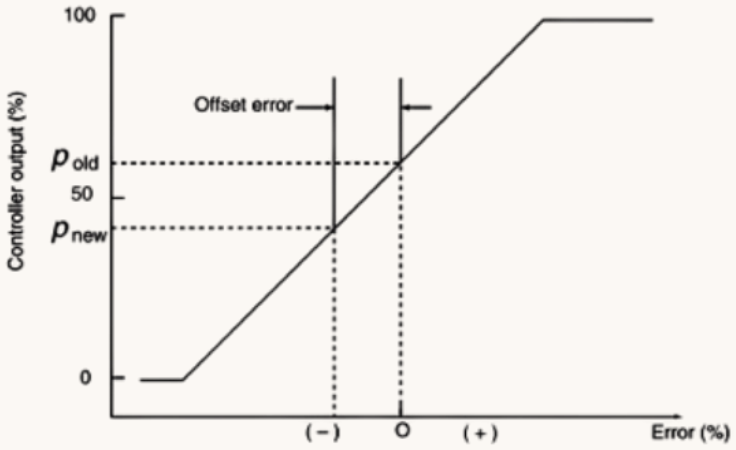
The proportional band is dependent on the gain. A high gain means large response to an error, but also a narrow error band within which the output is not saturated. In general, the proportional band is defined by the equation

$$PB = \frac{100}{K_P}$$

A plot of the proportional mode output versus error of this control mode is shown in Figure below. In this case, *P* has been set to 50% and two different gains have been used.



Offset: An important characteristic of the proportional control mode is that it produces a *permanent residual error* in the operating point of the controlled variable when a change in load occurs. This error is referred to as offset. It can be minimized by a larger constant, , which also reduces the proportional band. To see how offset occurs, consider a system under nominal load with the controller at 50% and the error zero, as shown in Figure below.



Summary:

- If the error is zero, the output is a constant equal to P_0.
- If there is error, for every 1% of error, a correction of K_p percent is added to or subtracted from P_0, depending on the sign of the error.
- There is a band of error about zero of magnitude *PB* within which the output is not saturated at 0% or 100%.

Integral-Control Mode:

Integral action is provided by summing the error over time, multiplying that sum by a gain, and adding the result to the present controller output. This mode is represented by an integral equation

$$p(t) = K_I \int_0^t e_P \, dt + p(0)$$

where *p(0)* is the controller output when the integral action starts. The gain K_I expresses how much controller output in percent is needed for every percent-time accumulation of error.

It can be summarized as:

- If the error is zero, the output stays fixed at a value equal to what it was when the error went to zero.

- If the error is not zero, the output will begin to increase or decrease at a rate of K_I percent/second for every 1% of error.
- The integral gain, K_I, is often represented by the inverse, which is called the *integral time*, or the *reset action*, $T_I = 1/K_I$.

Derivative-Control Mode:

Derivation controller action responds to the rate at which the error is changing— that is, the derivative of the error. Appropriately, the equation for this mode is given by the expression

$$p(t) = K_D \frac{de_p}{dt}$$

where the gain, K_D, tells us by how much percent to change the controller output for every percent-per-second rate of change of error. Derivative action is not used alone because it provides no output when the error is constant. Derivative controller action is also called *rate action* and *anticipatory control*.

The derivative mode can be summarized as:

- If the error is zero, the mode provides no output.
- If the error is constant in time, the mode provides no output.
- If the error is changing in time, the mode contributes an output of percent for every 1%-per-second rate of change of error.
- For direct action, a positive rate of change of error produces a positive derivative mode output.

Proportional-Integral Control (PI):

This is a control mode that results from a combination of the proportional mode and the integral mode. The analytic expression for this control process is found from a series combination of proportional and integral modes as given below:

$$p = K_P e_p + K_P K_I \int_0^t e_p \, dt + p_I(0)$$

where, $P_I(0)$ = integral term value at t = 0 (initial value)

Proportional-Derivative Control Mode (PD):

The analytic expression for this mode is found from a combination of proportional and derivative mode as given below:

$$p = K_P e_p + K_P K_D \frac{de_p}{dt} + p_0$$

Three-Mode Controller (PID):

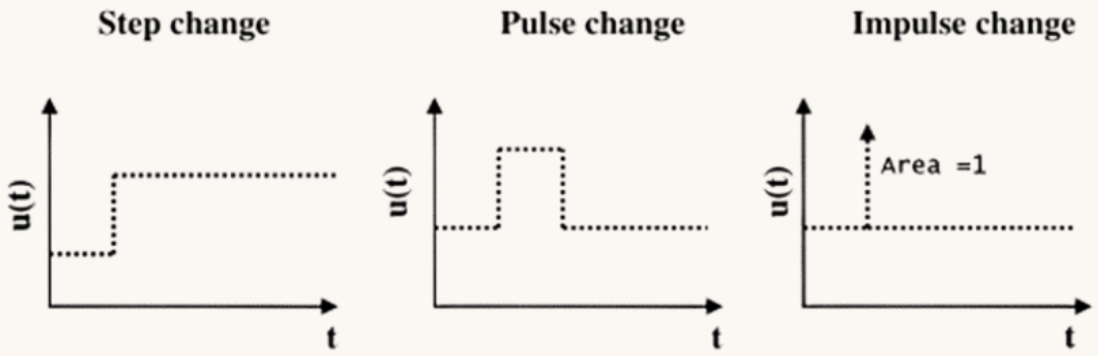
One of the most powerful but complex controller mode operations combines the proportional, integral, and derivative modes. This system can be used for virtually any process condition. The analytic expression is

$$p = K_P e_p + K_P K_I \int_0^t e_p dt + K_P K_D \frac{de_p}{dt} + p_I(0)$$

2.4 Test Inputs and Responses

Introduction:

To systematically characterize the transient response of an output to a change in the input, the input change is usually standardized to a step change, a pulse change or an impulse change. These standard input changes are depicted in Figure below. A step change in the input, the simplest input change pattern, is normally used to characterize the process dynamics.

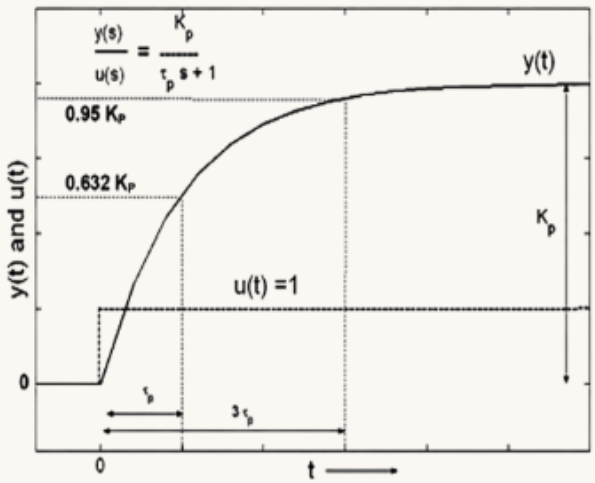


Basic Response Types:

The dynamics of every process are. Even so, the variety of transient responses can be characterized as an appropriate combination of one or more basic response types as described below.

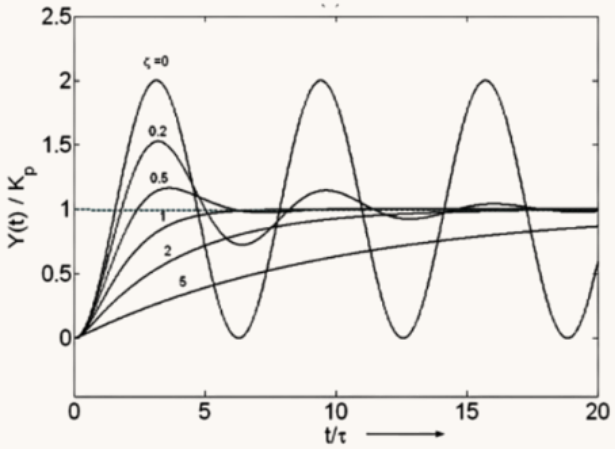
First Order Lag:

The first order lag is the simplest transient response where the output immediately responds to a step change in the input as given in Figure below. The ratio of the change in the output to the change in the input is referred to as the process gain, Kp. The time it takes for the output to reach 63.2% of its final value corresponds to the first order time constant τ_p. The output reaches ~95% of its final value in 3 time constants and reaches ~98% of its final value in 5 time constants. The time taken to reach $5\tau_p$ is called as settling time.



Second Order Response:

Sometimes, a step change in the input causes the output to oscillate before settling at the final steady state. The simplest such response corresponds to a second order underdamped system. The damping coefficient, ξ, can be used to characterize all second order responses – overdamped (ξ > 1), critically damped (ξ = 1) and underdamped (ξ < 1). The second order response is shown in Figure below.

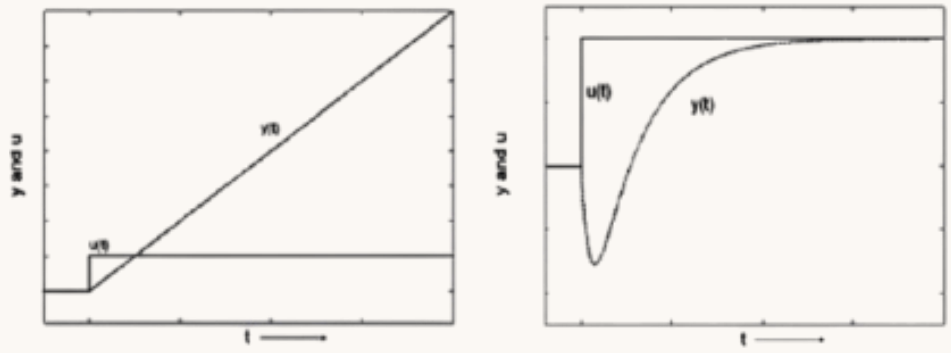


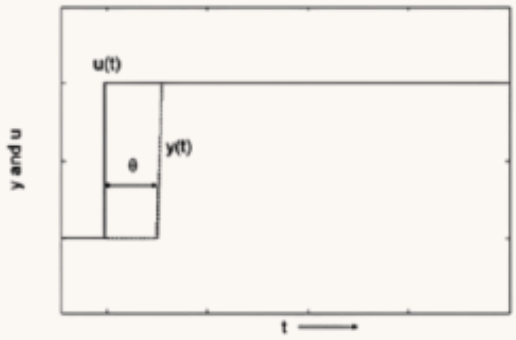
To gain an appreciation of the impact of damping coefficient on the transient response, Table below reports the ratio of the second overshoot to the first overshoot for different values of ξ. A quarter decay ratio is observed for a damping coefficient of 0.218. Sustained oscillations (decay ratio = 1) are observed for a damping coefficient of 0. As ξ increases to 1, the overshoot in the output disappears.

Damping Coefficient, ζ	0	0.05	0.1	0.2	0.218	0.4	0.6	1
Decay ratio	1.000	0.730	0.532	0.277	0.250	0.064	0.009	0.000

Other Common Response Types:

Other types of responses include the pure integrator, the inverse response and the pure dead-time. The transient response to a unit step change can be seen in Figure below and are self-explanatory.





The most common example of a pure integrator is the response of the tank level to change in the inlet / outlet feed rate. Unless the inlet and outlet flows are perfectly equal, the tank level is either rising or falling in direct proportion to the mismatch in the flows. The level in a tank is thus non self-regulating with respect to the connected flows. A controller must be used to stabilize all such non self-regulating process variables.

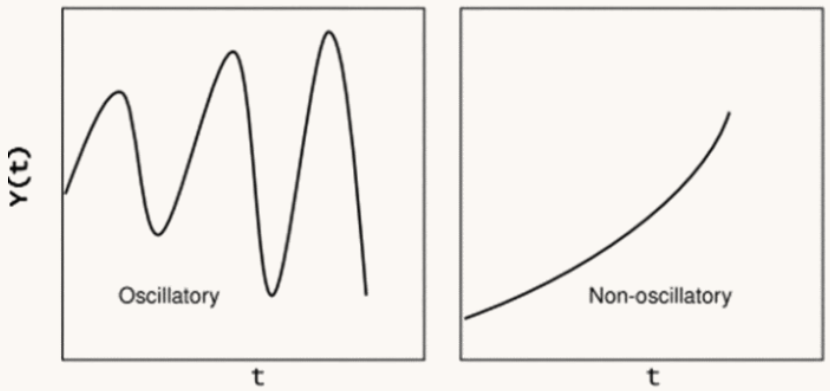
Dead time is very common in chemical processing systems and is due to transportation delay.

A very common example of the inverse response is the response of the liquid level in a boiler to a change in the heating duty. As the heating duty is increased, the vapour volume entrapped in the liquid increases causing the liquid interface level to rise initially. Over longer duration, the level of course reduces since more liquid is being vaporized. Dead time and inverse response can create control difficulties.

Unstable Systems:

Some systems may be inherently unstable. Unstable transient responses are shown in Figure below. The unstable response may be non-oscillatory or oscillatory as in the Figure.

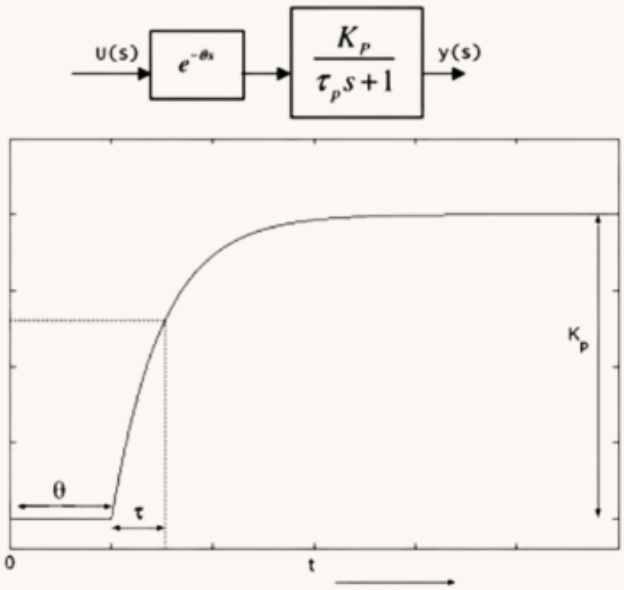
Reactor temperature runaway is an example of an unstable process. A control system must be used to stabilize an inherently unstable system.



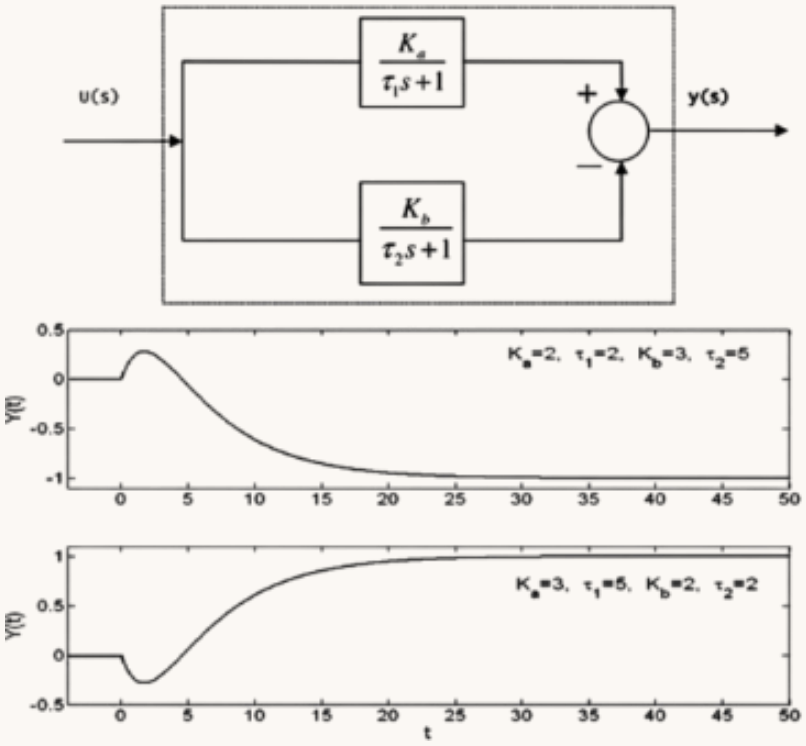
Combination of Basic Responses:

Any transient response can be reasonably represented as a combination of the above basic response types. One such combination is the first order lag plus dead time (FOPDT) that has been found to represent the

transient response of many chemical processing systems very well. The response is illustrated in Figure below.



Another example of such a combination is the inverse response which can be represented by the parallel combination of two first order lags. One of the lags has a small gain and a small time constant (ie a fast response) while the other lag has a gain of larger magnitude and opposite sign with a much larger time constant (i.e. a slow response in the opposite direction). Figure below illustrates this concept.



References:

http://nptel.ac.in/courses/103104049/2

2.5 Tuning Techniques

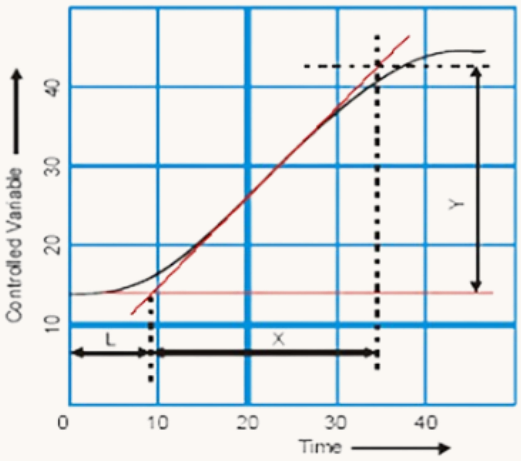
The three-mode controller (PID) is the most common feedback controller used in industrial control. The method of determination of the optimum mode gains, depending on the nature and complexity of the process is known as loop tuning.

The three parameters should be selected to meet a set of defined goals. These goals typically require a plant response with minimum steady state error, insensitivity to load disturbances and an acceptable transient response to set point changes and disturbances.

In practice the choice of proportional band, integral time and derivative time is a compromise between the set point tracking and disturbances. If a mathematical model of the process is known, selecting the controller parameters is relatively simple. But in many industrial applications, a reliable mathematical model is not available or it is difficult to determine. So empirical rules are developed for PID tuning which do not require mathematical model. A widely used set of rules is proposed by Ziegler- Nichols by open loop method and close loop method in 1942.

2.5.1 Open loop tuning (Process reaction method)

In open loop method the process is assumed to be model of first order. The step response i.e. process reaction curve, allows to obtain the approximate values of P, I and D parameters. With the feedback loop open, a step response is applied to manipulated variable and the values of P, I and D are estimated.



Where

Slope R: Slope of line drawn tangent to the point of inflection.

$$R = \frac{\%ChangeinVariable}{time(\min)} = \frac{Y}{X}$$

Dead time L: Time between the step change and the point where tangent line crosses the initial value of the controlled variable (in min.)

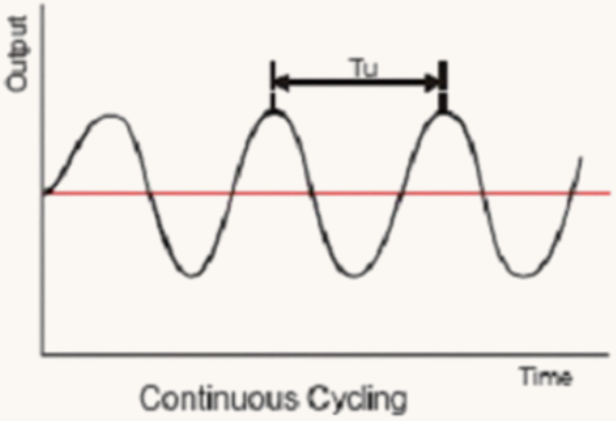
P = Step change applied in %

Using these parameters, the empirical equations are used to predict the controller settings for a decay ratio of 1/4. For P, PI and PID controller the parameters are calculated as follows.

Mode	Proportional band (in %)	Integral time (in Min)	Derivative time (in Min)
P	100RL/ΔP		
P+I	110RL/ ΔP	L/0.3	
P+I+D	83RL/ ΔP	L/0.5	0.5L

2.5.2 Closed loop tuning (Ultimate gain method)

This method is also called as ultimate gain method. The term ultimate was attached to this method because its use requires the determination of the ultimate proportional band and ultimate period. The ultimate proportional band, PBu is the minimum allowable value of proportional band (for a controller with only proportional mode) for which the system continuously oscillates at constant amplitude. The ultimate period, Tu is the period of response with the proportional band set to its ultimate value. To determine the ultimate proportional band and ultimate period the proportional band of the controller (with all integral and derivative action turned off) is gradually reduced until the process cycles continuously.



The process is placed in the closed loop with a proportional controller. The Proportional band is decreased until the process goes to continuous oscillations. The corresponding value of proportional band is called as ultimate proportional band PBu and the period of oscillation is called the ultimate period Tu. The PID controller parameters are selected from the following table.

Mode	Proportional	Integral	Derivative
P	2 PBu		
P+I	2.2 PBu	Tu/1.2	
P+I+D	1.65 PBu	0.5 Tu	Tu/8.0

2.6 Stability Analysis

Laplace transform techniques are used to express mathematical models in terms of transfer functions in order to calculate transient response. An alternative way to interpret the transfer function model, referred to as the complex transfer function or the frequency response. It shows how the output response characteristics depend upon the frequency of the input signal. This is the origin of the term, frequency response.

Stability analysis by using bode plots:

Frequency response may be plotted in three ways:

- The rectangular plot with amplitude ratio verses frequency and phase lag verses frequency.
- A polar plot with magnitude and phase shown in vector form with frequency as a parameter.
- A phase margin plot with magnitude shown verses a function of phase with frequency as a parameter.

If the system is linear, the output signal for sine wave input must be a sine wave having the same frequency as the input. Assume that the input to the first order system is suddenly switched from a constant input to a sine wave input. The initial slope of the sine wave input signal may be approximated by a ramp input or even exponential input.

If the input frequency is very low, period of the sine wave will be very long. Stated in another way, period is very much larger than the system time constant. Under this condition the output will have sufficient time to approach the condition, which is nearly in equilibrium with the input signal. Thus, at low frequencies both magnitude and phase angle of the output signal will be close to magnitude and phase angle of the input signal. However, as the frequency increases period decreases and the system do not have sufficient time to achieve the equilibrium status between input and output. Under these conditions output magnitude will decrease, and output phase angle will fall behind or lag the input phase angle. Thus, it is evident that any solution must show a decrease in ratio of output to input magnitude as frequency increases.

Magnitude and phase angle calculation:

Let,

M Magnitude ratio

A1 Input amplitude (%)

A2 Output amplitude (%)

Ø Phase angle (degrees)

X lag (seconds)

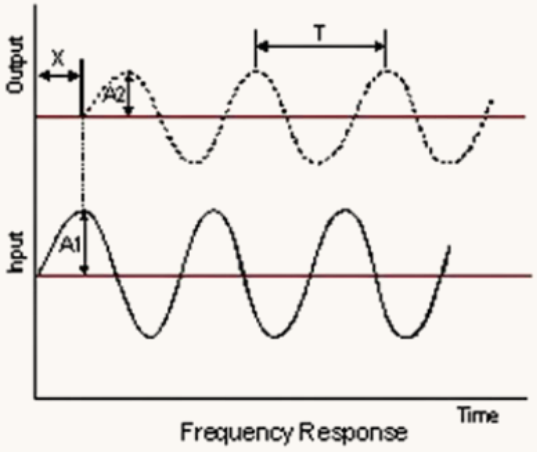
T Period (seconds)

f frequency (cycles per seconds) = 1/T.

Then,

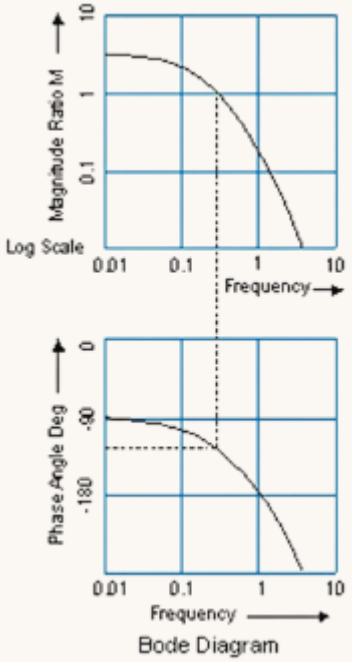
M = A2/A1

ø = (x/T) x 360



BODE diagram

In Bode diagram the magnitude ratio verses frequency is plotted on log-log coordinates and the phase angle verses frequency on semi-log coordinates.



Stability criteria

- A system is stable if the phase lag is less than 180 Deg. at the frequency for which the gain is unity (one).
- A system is stable if the gain is less than one (unity) at the frequency for which the phase lag is 180 Deg.

2.7 Mathematical Model

First Principle Model:

When flow occurs from a tank discharging under its own head or pressure, the flow is found by the following equation:

$$q_0 = KA\sqrt{2gh}$$

Where q_0 is the flow rate (ft^3/s), K is a flow coefficient, A is the area of the discharge orifice (ft^2), g is gravitation constant (ft/s^2), and h is pressure head of liquid (ft).

We can define hydraulic resistance (R) to flow as follows:

$$R \equiv \frac{Potential}{Flow} = \frac{h}{q_0}$$

The transient mass balance equation can be given as

Mass flow-in – Mass flow-out = Rate of accumulation of mass in the tank

In terms of variables used in this analysis, the mass balance equation becomes

$$\rho q_i(t) - \rho q_0(t) = \frac{d(\rho V)}{dt} = \frac{d(\rho A h)}{dt} \Rightarrow q_i(t) - q_0(t) = A\frac{dh}{dt}$$

Rearranging and eliminating q_0, yields

$$Rq_i(t) = RA\frac{dh}{dt} + h$$

Taking Laplace transform of the above equation,

$$\frac{H(s)}{Q_i(s)} = \frac{R}{(RAs+1)}$$

Empirical Model:

In empirical model building, models are determined by making small changes in the input variable(s) about a nominal operating condition. The resulting dynamic response is used to determine the model. This general procedure is essentially an experimental linearization of the process that is valid for some region about the nominal conditions.

The Process Reaction Curve Method:

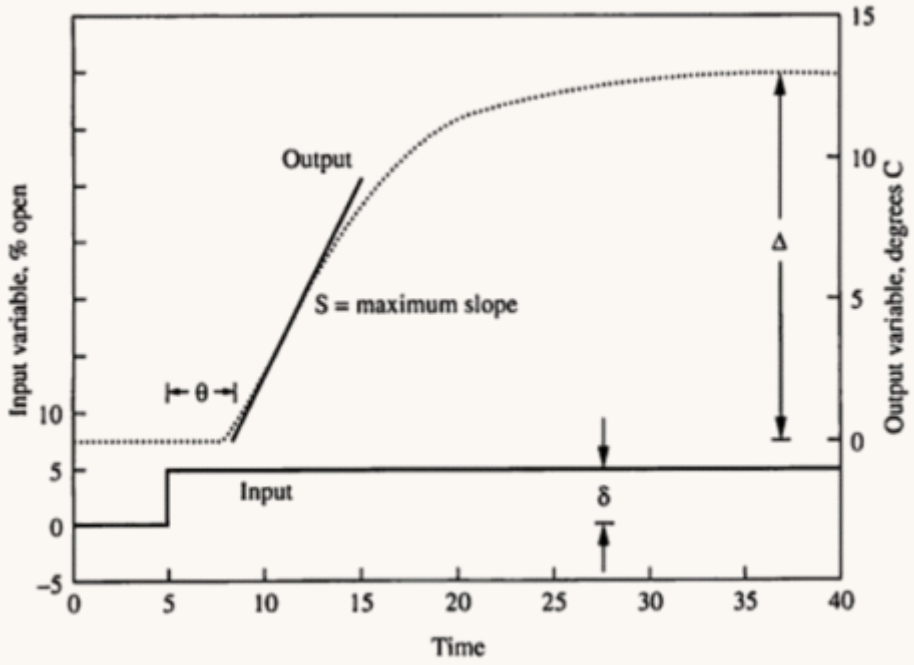
The process reaction curve method involves the following four actions:

- Allow the process to reach steady state.
- Introduce a single step change in the input variable.
- Collect input and output response data until the process again reaches steady state.
- Perform the graphical process reaction curve calculations.

The graphical calculations determine the parameters for a first-order-with-dead-time model: the process reaction curve is restricted to this model. The form of the model is as follows, with X(s) denoting the input and Y(s) denoting the output, both expressed in deviation variables:

$$\frac{Y(s)}{X(s)} = \frac{K_p e^{-\theta s}}{\tau s + 1}$$

This technique, adapted from Ziegler and Nichols (1942), uses the graphical calculations shown in Figure.



Where,

S - Maximum Slope

Δ - magnitude of the steady-state change in the output

- magnitude of the input change

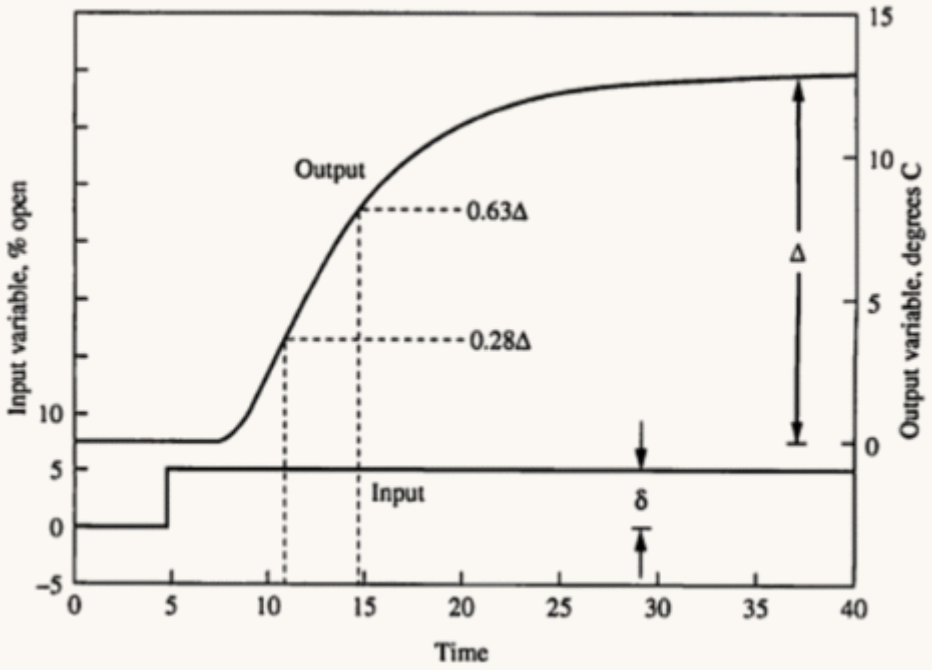
The model parameters can be calculated as

$$K_p = \frac{\Delta}{\delta}$$

$$\tau = \frac{\Delta}{S}$$

Two Point Method:

This method uses the graphical calculations shown in Figure. The intermediate values determined from the graph are the magnitude of the input change, δ; the magnitude of the steady-state change in the output, Δ; and the times at which the output reaches 28 and 63 percent of its final value.



The values from the plot can be related to the model parameters using the general expression in equation below. Any two values of time can be selected to determine the unknown parameters, θ and τ. The typical times are selected where the transient response is changing rapidly so that the model parameters can be accurately determined in spite of measurement noise (Smith, 1972). The expressions are

$$Y(\theta + \tau) = \Delta(1 - e^{-1}) = 0.632\Delta$$
$$Y(\theta + \tau/3) = \Delta(1 - e^{-1/3}) = 0.283\Delta$$

The values of time at which the output reaches 28.3 and 63.2 percent of its final value are used to calculate the model parameters.

$$t_{28\%} = \theta + \frac{\tau}{3} \qquad t_{63\%} = \theta + \tau$$
$$\tau = 1.5(t_{63\%} - t_{28\%}) \qquad \theta = t_{63\%} - \tau$$

Ideally, both techniques should give representative models; however, Method I require the engineer to find a slope (i.e., a derivative) of a measured signal. Because of the difficulty in evaluating the slope, especially when the signal has high-frequency noise, Method I typically have larger errors in the parameter estimates; thus, Method II is preferred.

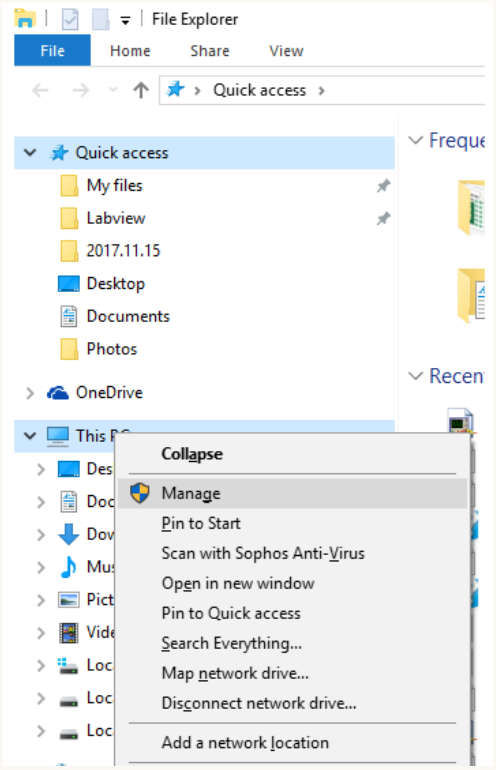
References:

1. Katsuhiko Ogata, “Modern Control Engineering”, 5th Edition.
2. Thomas E Marlin, “Process Control: Designing Processes and Control Systems for Dynamic Performance”, 2nd Edition.

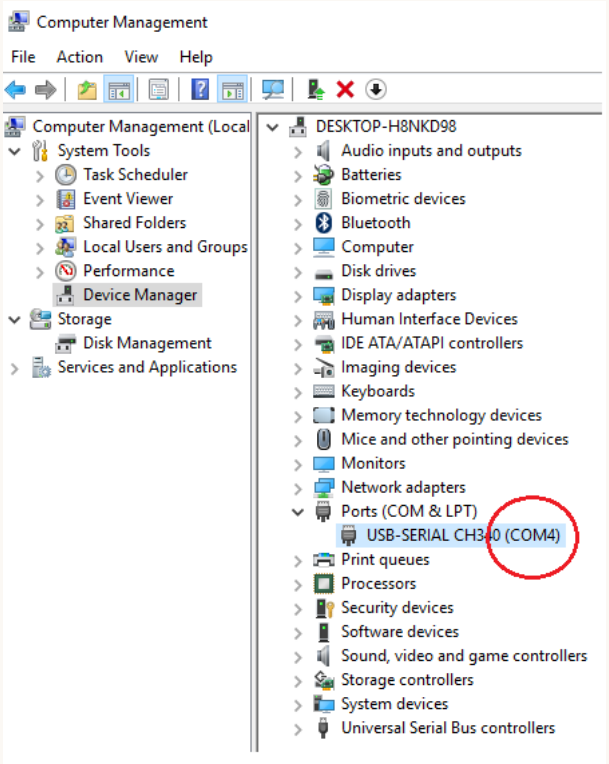
3. Software Interface & Features

3.1 Device Connection

1. Open 'Mange' by right clicking 'My Computer' or 'This PC' by opening Windows Explorer or from Desktop.



2. Select Device Manager >> Ports (COM & LPT). Locate your device port No. (COM4 as shown below).



3. If no device is visible ensure your communication cable is connected properly, check for polarity or driver software. Mail to contact@advancedcontrol.in for further assistance.

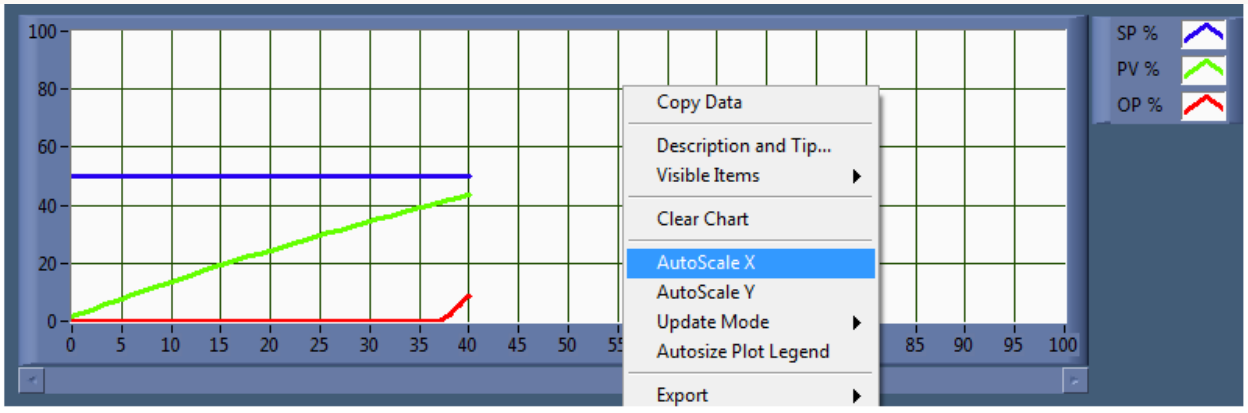
3.2 Features

The software is designed to be user friendly and enhanced for a wide variety of options with single click. In running mode, the following options can be explored and used. The features of the software are listed below:

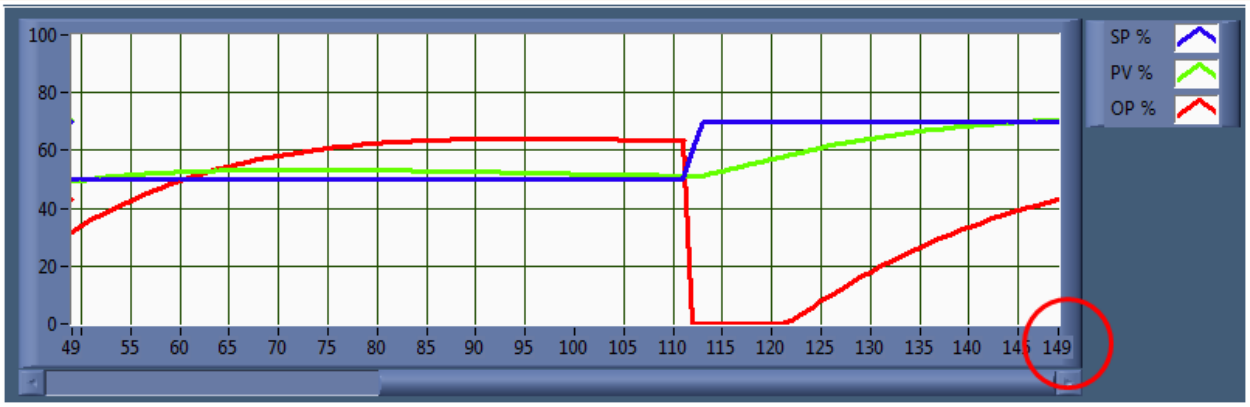
- No loss of data from the beginning to the end
- All the data are stored internally using the graph and can be extracted at any time
- 4k memory allocation which allows the user to run the equipment hours together without any interruption
- Graph axis can be altered as per requirement to get a complete view of the experiment performed
- Portion of data can also be logged and exported with current time.
- Excel, text and image format support

1. Graph axis control:

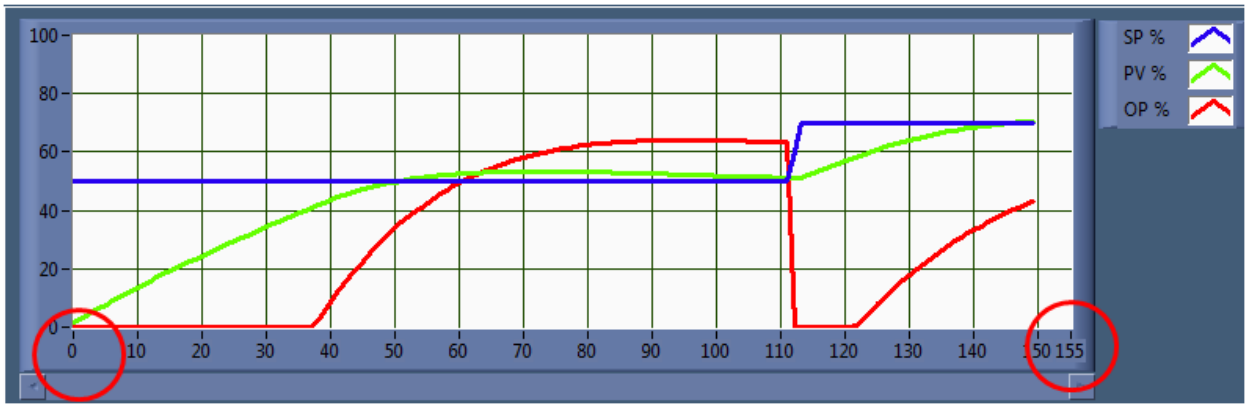
   a. Right click on the graph and **disable** AutoScale X as shown below:



   b. Double click the left bottom axis value and change it to a new value. Then double click the right bottom axis value and change it to 0.

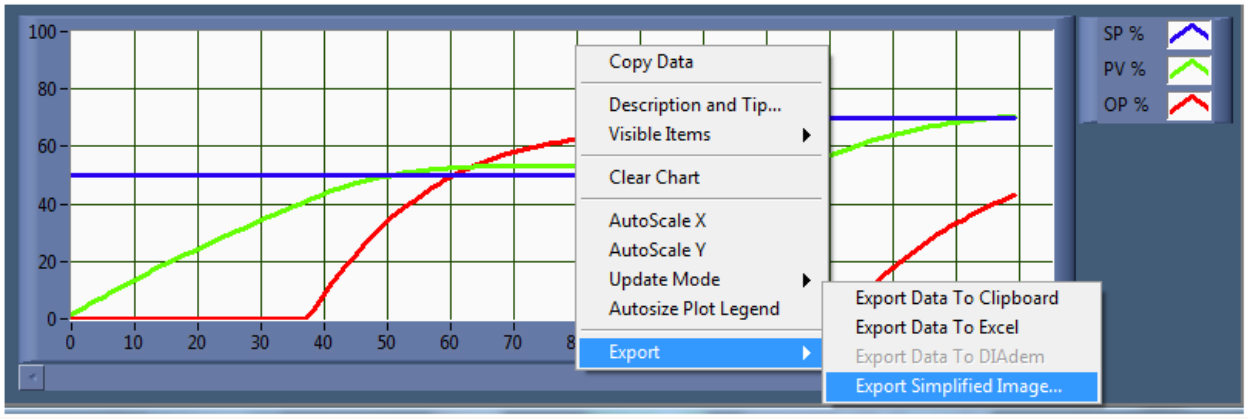


   c. The axis will be automatically readjusted to the new value as shown below and the entire plot can be viewed.



2. Export Graph:

    a. The graph can be exported to a simplified image by right clicking on the graph as shown below:



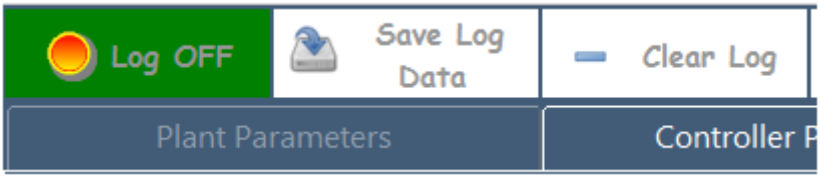
    b. In the above case the graph background will be changed to white and scale will be converted to auto scale.

    c. The graph image as it is can be exported to an image file by clicking on the 'Export Graph' button on the menu bar.

    d. Graph data can be exported to either excel or clipboard by clicking appropriate option as shown above.

3. Log data:

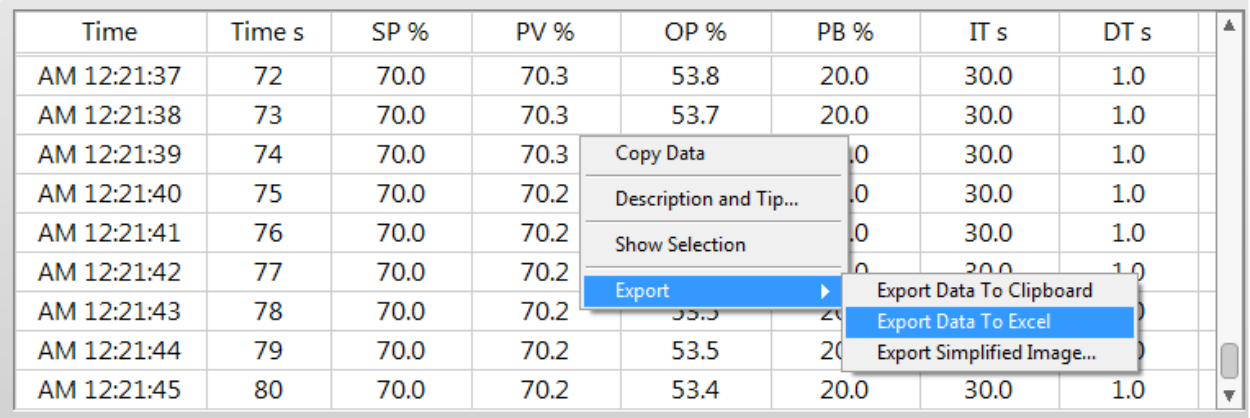
- The data can be logged from any point of time to any other time value using 'Log ON' button. Use the same button to stop log. An indicator will be turned on to indicate the log process is ON.



    a. The values will be displayed in the log table. Use 'Clear Log' to clear the data from Log table.

Time	Time s	SP %	PV %	OP %	PB %	IT s	DT s
AM 12:20:53	28	70.0	71.6	54.1	20.0	30.0	1.0
AM 12:20:54	29	70.0	71.6	54.2	20.0	30.0	1.0
AM 12:20:55	30	70.0	71.5	54.3	20.0	30.0	1.0
AM 12:20:56	31	70.0	71.5	54.4	20.0	30.0	1.0
AM 12:20:57	32	70.0	71.5	54.5	20.0	30.0	1.0
AM 12:20:58	33	70.0	71.5	54.6	20.0	30.0	1.0
AM 12:20:59	34	70.0	71.4	54.6	20.0	30.0	1.0
AM 12:21:00	35	70.0	71.4	54.7	20.0	30.0	1.0
AM 12:21:01	36	70.0	71.4	54.7	20.0	30.0	1.0

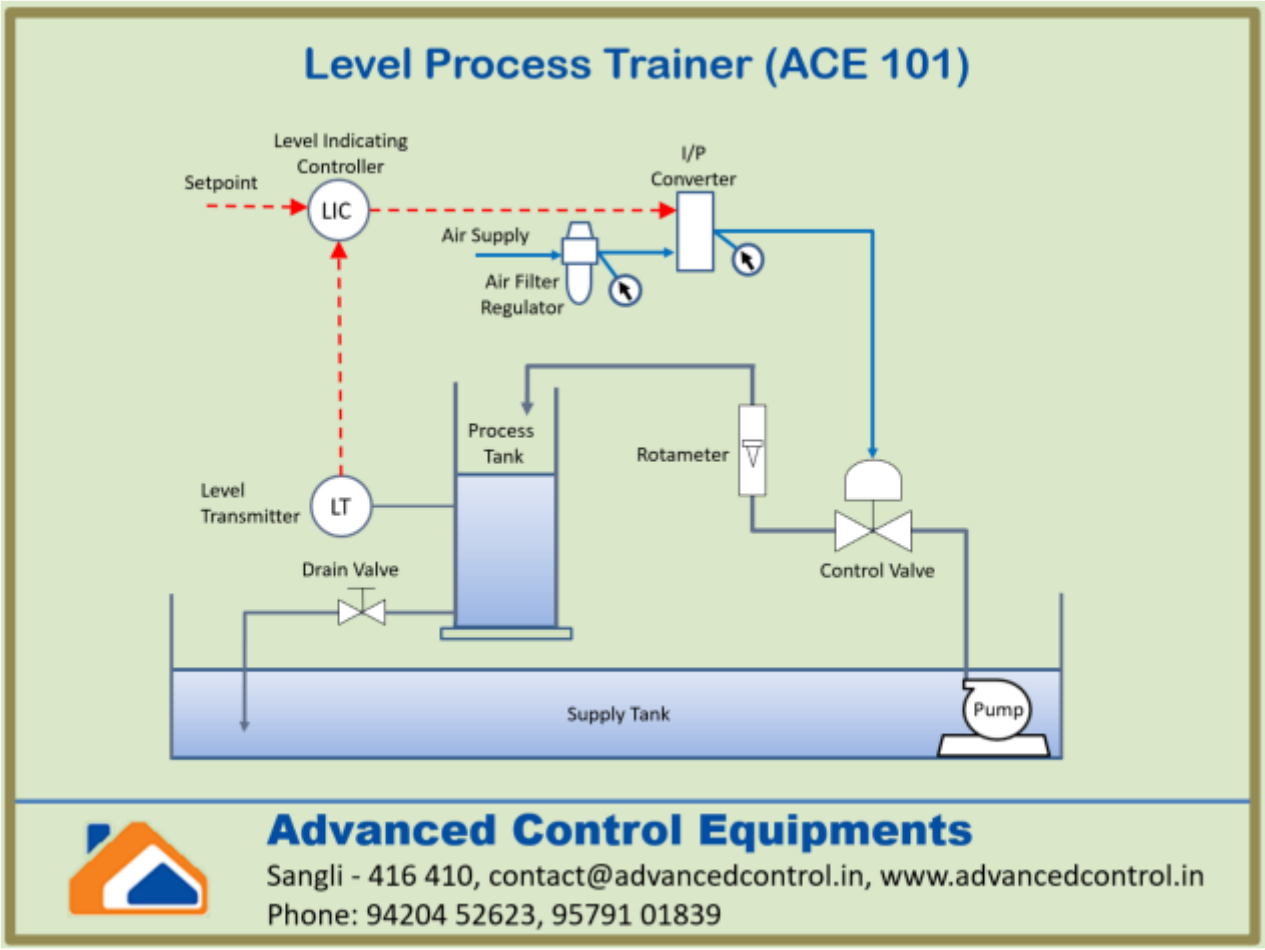
b. The data can be exported by right click on the Log table >> Export >> Export Data To Excel or Export Data to Clipboard



c. Save Log Data will be used to export the data directly to excel.

4. Experiments

The level is controlled by a pneumatically actuated control valve that operates on a standard 3–15 psi signal. Because the controller generates a 4–20 mA electrical output, a current-to-pressure (I/P) converter is employed to transform this electrical signal into the corresponding pneumatic pressure required by the valve. The level is measured and regulated by manipulating the inlet flow through a digital indicating controller, which allows adjustment of set points and tuning of control parameters. Open loop, On-Off and PID control-based experiments can be performed using this trainer. Refer the P&I diagram before performing the experiment.



4.1 Study of Level Transmitter Characteristics

General Procedure:

1. Turn on the electrical supply and switch on both the mains and the pump.
2. Ensure the compressor is switched off, or close the air supply valve so that no air is supplied to the trainer.
3. Adjust the rotameter valve to maintain the flow rate close to 100 LPH.
4. Regulate the drain valve of the level tank so that the tank level remains between 95% and 100%.
5. Close the rotameter valve completely and then gradually increase the flow rate from 0 LPH to 100 LPH in steps of 10 LPH.
6. For each step, observe and record the flow rate indicated by the rotameter (LPH), the actual tank level (cm) from the graduated scale, and the measured level (%) displayed on the controller.
7. **Note:** 0-25 cm of tank height is calibrated to 0 to 100%. The conversion to Actual Level (%) can be given as Actual Level (%) = Actual Level (cm) * 4.

Observations:

- Tabulate the observations as follows

Flow in LPH	Actual Level (cm)	Actual Level (%)	Measured Level (%)
0			
10			
20			
…			
100			

- Plot Actual Level (%) with Measured Level (%) and observe the deviation.
- Calculate the percentage of error (Mean Square Error).
- Do a upside down reading (100, 90 …10 LPH) and calculate the hysteresis.

4.2 Open Loop, On-Off & Closed Loop Experiments

The level process consists of a control valve with an I/P converter, a pressure transmitter (calibrated to level) fitted at the bottom of the process tank, and a rotameter for flow adjustment. The inflow to the tank is regulated by modulating the control valve through the I/P converter arrangement.

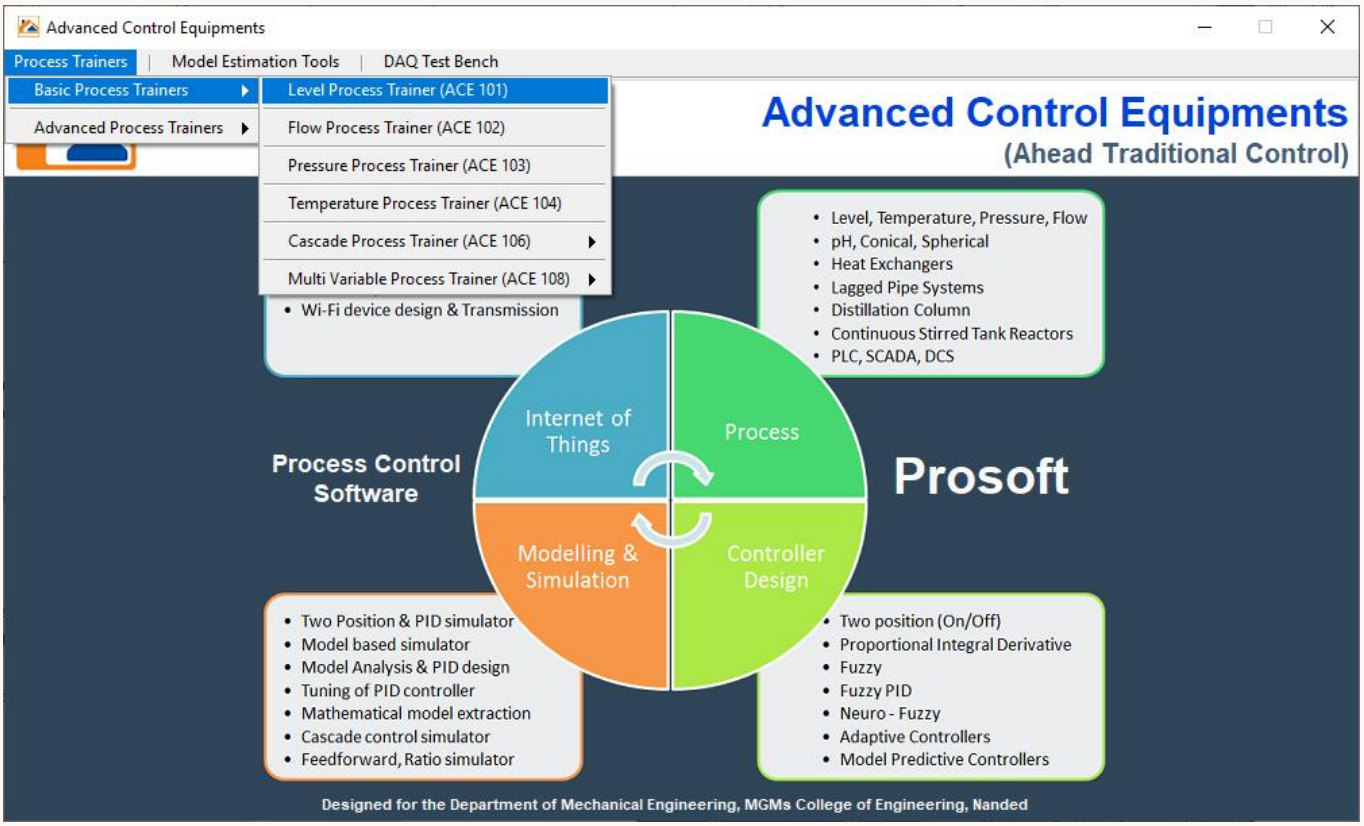
General Procedure:

Hardware:

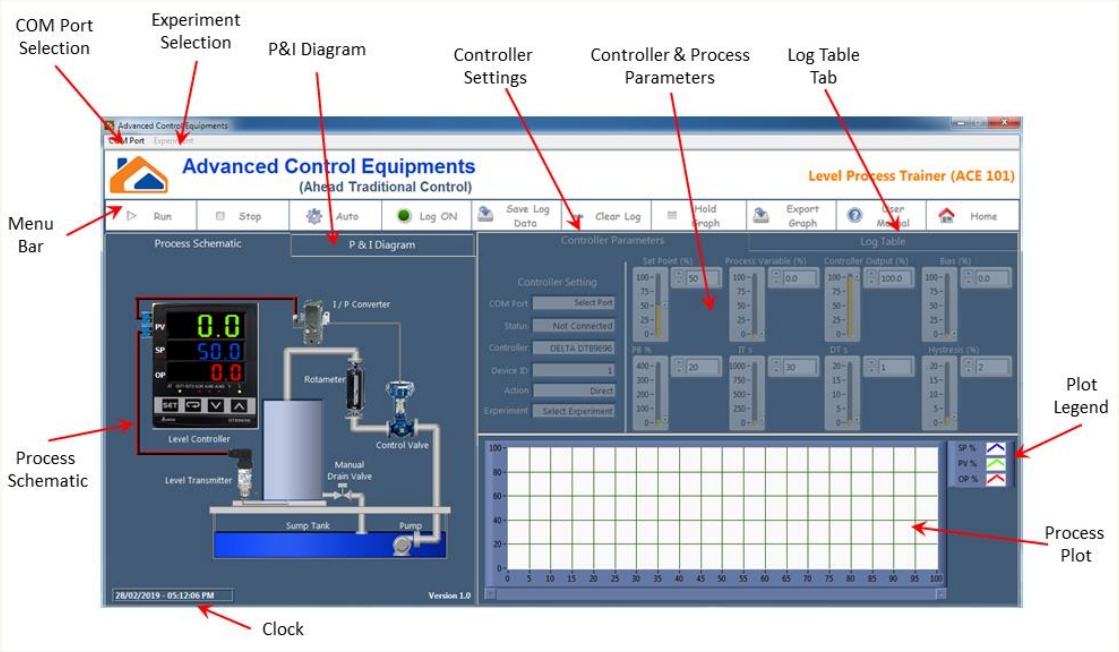
1. Turn on the electrical power supply and switch on both the mains and the pump.
2. Ensure the compressor is switched off, or close the air supply valve so that no air is supplied to the trainer.
3. Adjust the rotameter valve to maintain the flow rate close to 100 LPH.
4. Regulate the drain valve of the level tank so that the tank level is maintained between 95% and 100%.
5. Switch on the compressor or open the air supply valve and ensure that the air pressure supplied is above 25 psi.

Software:

1. Double click "ACE.exe" icon in desktop.
2. Select: Basic Process Trainers >> Level Process Trainer (ACE 101)



3. Wait for the GUI to open. Sample description of the software screen is given below.

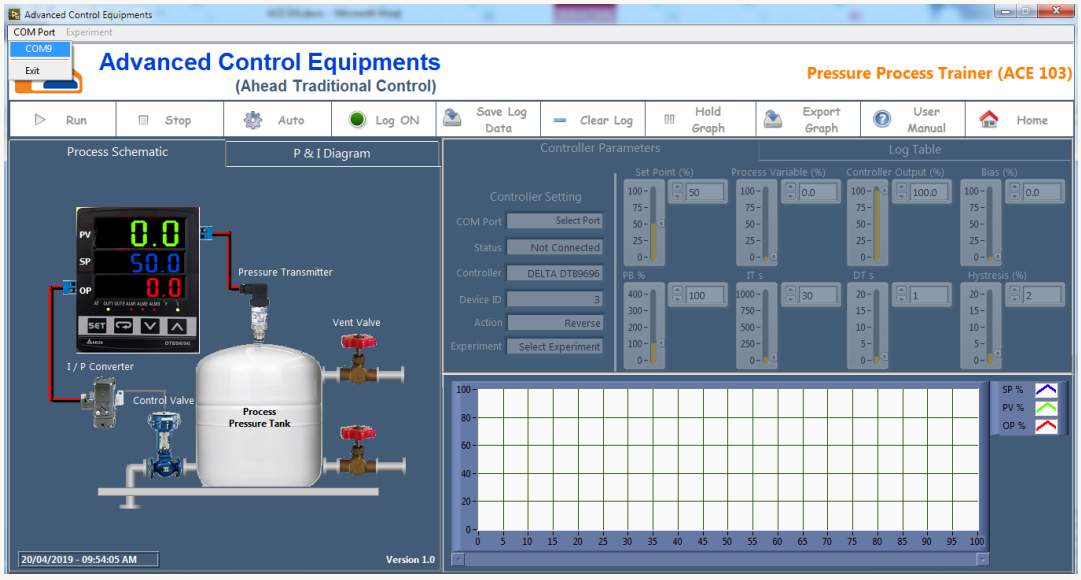


4. Select COM Port from the Menu Bar.

***Note:**

a. Enabled COM ports will be automatically identified and displayed here. If 'No Device' is displayed, check your communication cable. Also ensure your RS485 to USB converter is installed properly.

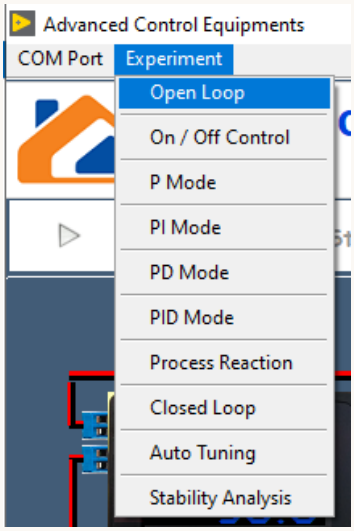
b. Go through the procedure to ensure device connection to establish communication between Delta DTB9696B controller with PC / Laptop.



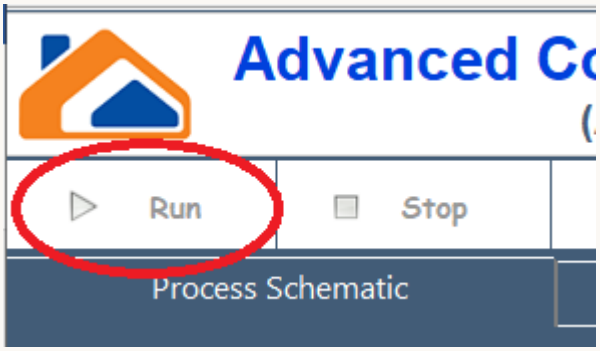
5. Once communication is established you will get a message as shown below. 'Experiment' menu will be enabled and 'COM Port' menu will be disabled.



6. Click 'OK' and select 'Experiment'.



7. Once experiment is selected, you will get a message "Experiment Selected. Press 'RUN' to continue". Click OK and Click 'Run' button.



4.2.1 Open Loop

1. Refer Software Interface & Features and General Procedure.
2. Establish communication and Select Open Loop from Experiment Menu.
3. Adjust the controller output to 100%. Note down steady state process value.
4. Apply the step change by -10% to controller output and wait for the level to reach the new steady state value. Note down the process value.
5. Repeat the above step until the controller output reaches to minimum i.e. 0% and for each change, note steady state process value.
6. Use 'Log on', 'Export', 'Clear Graph', 'Hold Graph' options as and when required. You can also change the graph axis scale. Click here for more details.
7. Press 'Stop' button to finish the experiment or to jump to some other experiment. Press Ok or Cancel button on the popup window.

Observations:

- Tabulate the observations as follows

Controller output in %	Process Value in %
0	
10	
20	
…	
100	

- From the above data, note the output required for maintaining the level at desired set points.

4.2.2 On/Off Control

1. Refer Software Interface & Features and General Procedure.
2. Establish communication and Select On / Off Control from Experiment Menu.
3. Change Hysteresis value to 5%. (Range 1-10%)
4. Change the values of the set point and observe the On-Off control operation.

Observations:

- Observe that if process value exceeds the set point and increases above the value of (0.5x Hysteresis), control valve is fully closed and if process value decreases below the set point by (0.5 x Hysteresis), the control valve opens fully i.e. controller operates like On/Off switch.

4.2.3 P Mode

1. Refer Software Interface & Features and General Procedure.
2. Establish communication and Select **P Mode** from Experiment Menu.
3. Keep the set point to 50%. Allow the process to stabilize.
4. Change output mode to Manual. Adjust output value so as to match the process value with set point and apply this output value as bias value to the controller.
5. Switch the controller to Auto mode. Adjust the proportional band to 100%.
6. Apply step change to set point (say 50% to 60%) and observe the response.
7. Decrease proportional band to half of the previous value. With each decrease, obtain a new response of the step change. Ensure that the set point changes are around the same operating point.
8. Using trial and error approach, find a value of proportional band so that the response to a step change has **at most one overshoot and one undershoot.**

Observations:

- Observe steady state error decreases as proportional band decreases.
- Observe the effect of very low proportional band values (system works in oscillatory mode).
- Observe the response of the system to load change. Load change can be given by slightly opening the drain valve for a shorter duration and retain the same position.
- Apply a setpoint change (say 50% to 60%) with the proportional band obtained using trial and error approach. Plot the responses.

4.2.4 PI Mode

1. Refer Software Interface & Features and General Procedure.
2. Establish communication and Select PI Mode from Experiment Menu.
3. Keep the set point to 50%. Allow the process to stabilize.
4. Set the proportional band estimated from Proportional control (P only) experiment and with integral time = 50.
5. Decrease the integral time by half the previous value.
6. Apply step change to set point (say 50% to 70%) and observe the response.
7. Keep the set point to 50%. Allow the process to stabilize.
8. Decrease the integral time again half of the previous value and observe the process response for step change.
9. Repeat the procedure until a satisfactory control is obtained with one overshoot and one undershoot and with very less offset.

Observations:

- Observe the effect of reducing integral time on the response of the process. Also note that the process may show very less or null offset as effect of integral action but with a sluggish response.
- Apply a setpoint change (say 50% to 70%) with the integral value obtained using trial and error approach. Observe the responses and the time taken to reach steady state.

4.2.5 PD Mode

1. Refer Software Interface & Features and General Procedure.
2. Establish communication and Select PD Mode from Experiment Menu.
3. Keep the set point to 50%. Allow the process to stabilize.
4. Set the proportional band estimated from Proportional control (P only) experiment and with derivative time=1.
5. Allow the process to reach at steady state.
6. Increase the derivative time by 1 sec.
7. Apply step change to set point (say 50% to 60%) and observe the response.
8. Increase the derivative time gradually (1 to 2 and 2 to 3) and observe the process response for step change.

Observations:

- Observe the effect of increasing derivative time. Also note that the process may show offset as effect of integral action is cut off.

4.2.6 PID Mode

1. Refer Software Interface & Features and General Procedure.
2. Establish communication and Select PID Mode from Experiment Menu.
3. Keep the set point to 50%. Allow the process to stabilize.
4. Change the proportional band to the value that estimated in proportional controller.
5. Set integral time and derivative time based on the responses in previous experiments.
6. Change the controller to Auto mode. Apply step change to set point (say 50% to 60%) and observe the response.
7. Change the proportional band, integral time, derivative time as per the requirement and observe the response of the process for step change for each change in setting.

Observations:

- Compare the steady state response of the PID controller with P, PI and PD controller obtained in the previous experiments.
- Apply a setpoint change (say 50% to 60%).
- Adjust the P, I and D values using trial and error approach to obtain a better response.
- Achieve a step response with at least one overshoot and one undershoot. Plot the response.
- Compare the response with previous experiment responses.

4.2.7 Open Loop Tuning

1. Refer Software Interface & Features and General Procedure.
2. Establish communication and Select Process Reaction from Experiment Menu.
3. The default controller output is 80%. Wait for the system to stabilize.
4. Start data logging.
5. Apply 60 % as controller output. Wait for the steady state. Stop data logging.
6. Plot the step response (Process reaction curve) from stored data or export from graph.
7. Find out the value of slope at the point of inflection and time lag.

PID Tuning:

- Calculate P I D settings for different modes using Open Loop Tuning explained earlier.
- Select PID Experiment. Set the PID values obtained from the calculations.
- Apply the step change & observe the response of the system.
- Allow the system to reach steady state and plot the response.

Observations:

- Observe the step response of the model using any standard tool like MATLAB and compare the response with experimental response.

4.2.8 Closed Loop Tuning

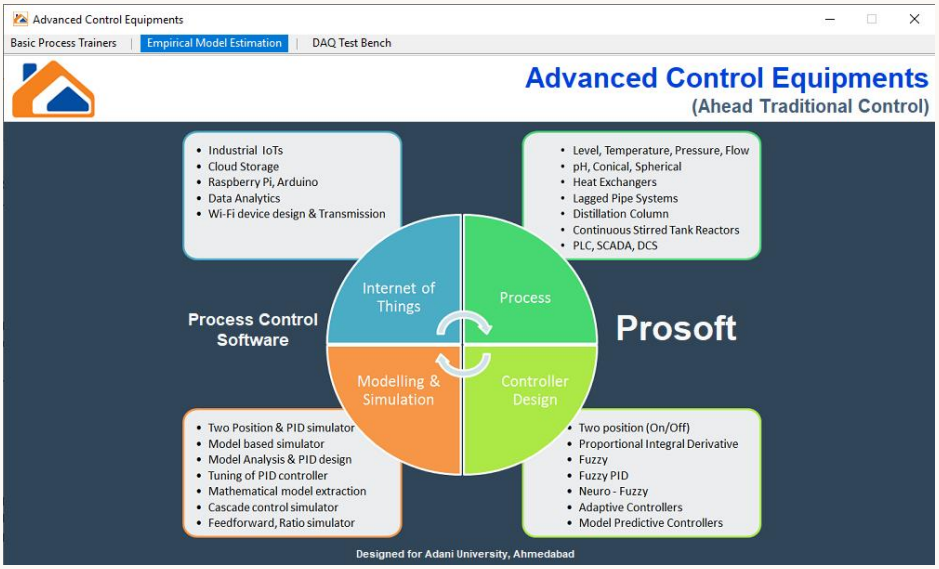
1. Refer Software Interface & Features and General Procedure.
2. Establish communication and Select Closed Loop from Experiment Menu.
3. Set the controller to manual mode and adjust the output so that the process value reaches to 50%.
4. Switch the controller to auto mode.
5. The default proportional band is 100%. Apply a step change to the set point and observe the process response.
6. Decrease the proportional band to half of the previous value and apply a step change to the set point and observe the process response.
7. Repeat the above procedure and find out correct value of proportional band for which the system just goes unstable i.e. continuous oscillations are observed in the output of controller.
8. Record the ultimate proportional band and ultimate period from the response.
9. Calculate the PID values from Closed loop method explained earlier.
10. Select PID Experiment. Set the PID values obtained from the calculations.
11. Apply a step change & observe the response of the system.
12. Allow the system to reach steady state and plot the response.

4.3 Empirical Model Estimation

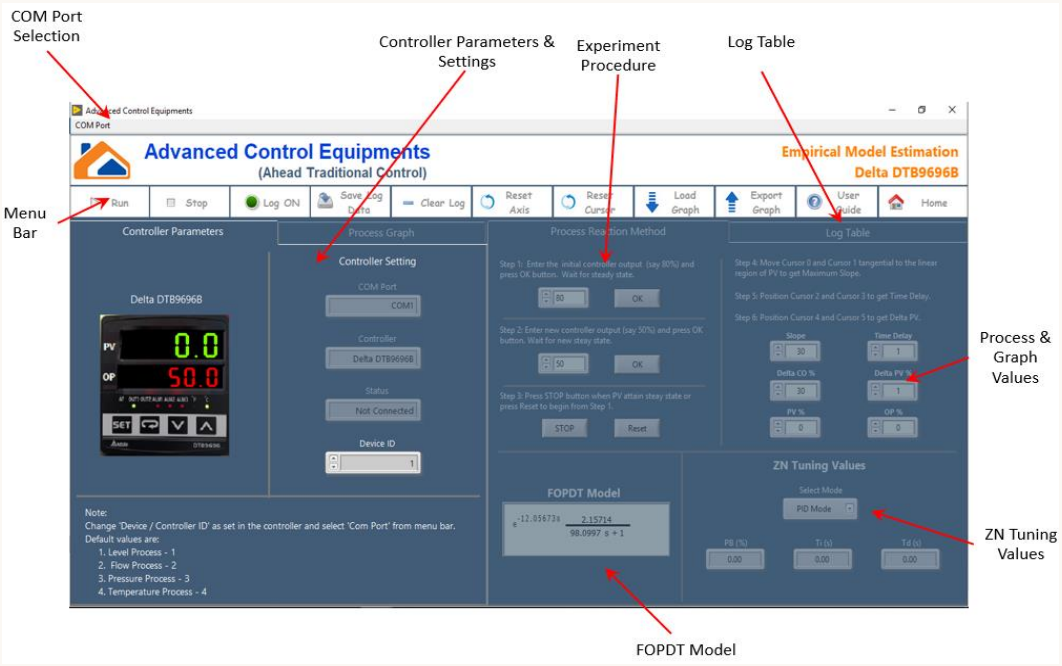
This tool aids to understand the procedure of model estimation using empirical (experimental) methods. A First Order Plus Dead Time (FOPDT) model can be estimated using Process Reaction Method using this tool. Refer Process Reaction Method from theory. This tool also provides facility to store the data in graph format and restore from stored file.

General Procedure:

1. Double click "Ace.exe" icon in desktop.
2. Select: Empirical Model Estimation tab.

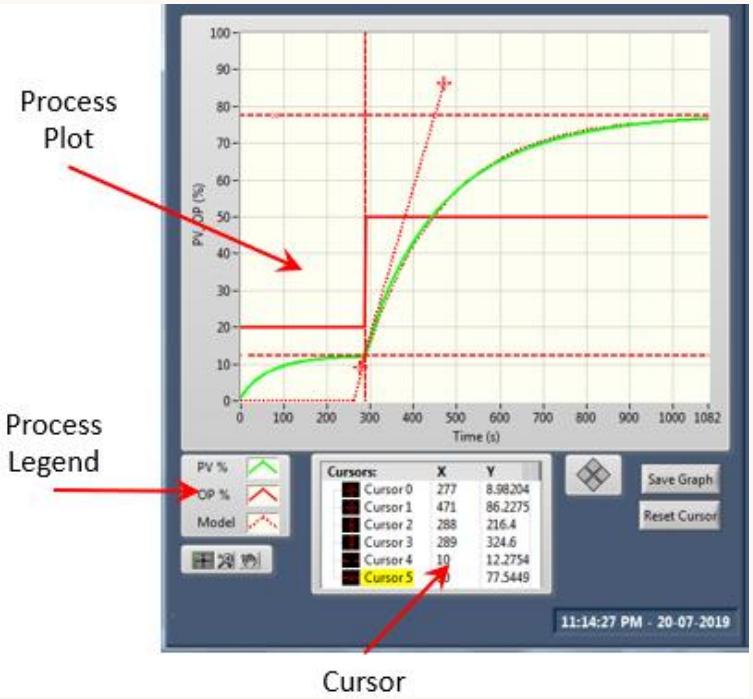


3. Wait for the GUI to open. The description of the page is shown below.



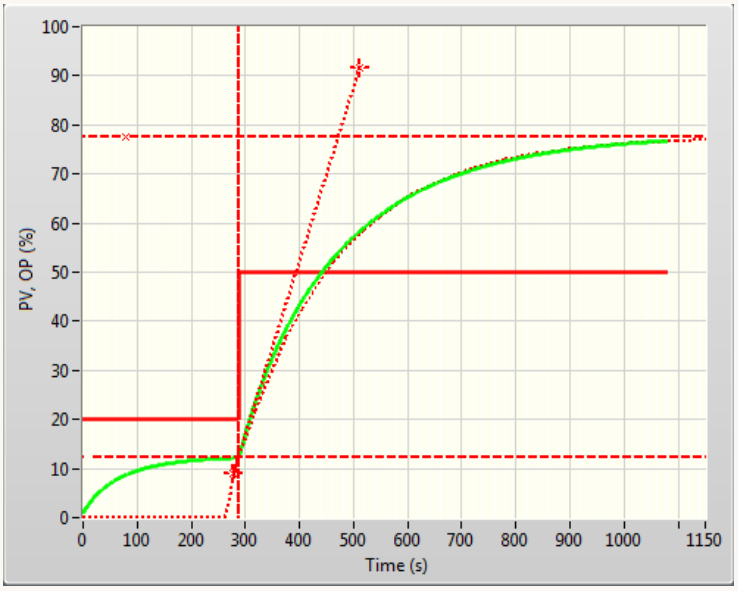
4. The experiment can be performed by following the instructions on the screen as listed below:
   - Select ‘Device ID’ (Flow Process with Delta DTB9696B controller the default value is 2.

- Select COM Port from the Menu Bar.
- *Note:
  - Enabled COM ports will be automatically identified and displayed here. If 'No Device' is displayed, check your communication cable. Also ensure your RS485 to USB converter is installed properly.
  - Go through the procedure to ensure device connection.
- Once communication is established you will get a message as 'Communication Established. Press RUN to continue'.
- Click RUN. Process Graph tab will be enabled. Graph tab will be enabled. Description of the tab is given below:



- Enter the initial controller output (say 20%) and press OK button. Wait for steady state. (Once if OK button is pressed, ‘Process Plot’ tab will be enabled. You can also scroll to ‘Process Schematic’. Observe the response on the process plot.)
- Press ‘STOP’ if you want to restart at any time.
- Once steady state is achieved, enter new controller output (say 50%) and press OK button. Wait for new steady state.
- Press STOP button when PV attain steady state.
- Once STOP button is pressed, cursors will be enabled in the process plot.
- Move Cursor 0 and Cursor 1 tangential to the linear region of PV to get Maximum Slope.
- Position Cursor 2 and Cursor 3 to get Time Delay. Use zoom option, if required.
- Position Cursor 4 and Cursor 5 to get Delta PV.
- The FOPDT model and ZN open loop tuned PID values will be displayed automatically.

5. A red dotted line represents the response of the model. Adjust cursors 0,1 and 4,5 to get a model close to the actual response.
6. A typical graph is shown below.



The following features can be used while doing the experiment.

1. Zoom option can be used to select the exact time delay (Fig a and b) as shown below.



Fig. (a)

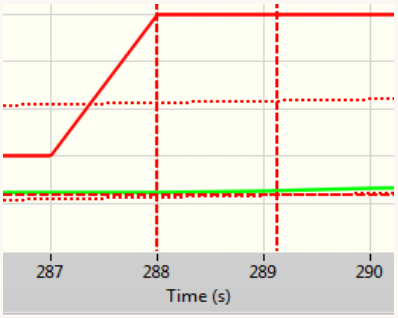
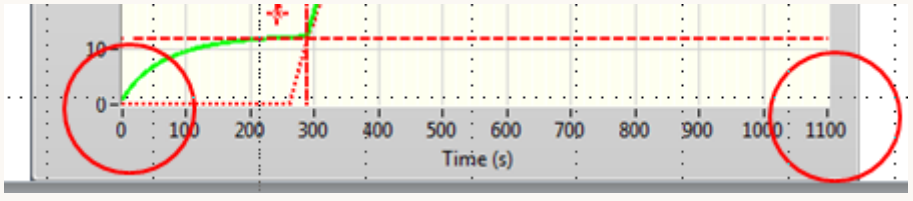


Fig. (b)

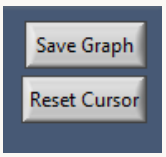


Fig. (c)

2. Select Cursor tool to adjust cursors as shown in fig (c). Zoom so wide and move cursors to get the exact time delay.
3. Adjust the time axis by double clicking the corner values as shown below.



4. Use ‘Reset Cursor’ if cursors are moved beyond the range.

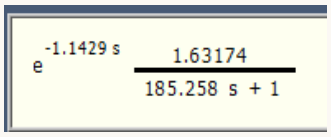


5. Use ‘Save Graph’ button to save the graph data. The graph will be saved in .gd format.

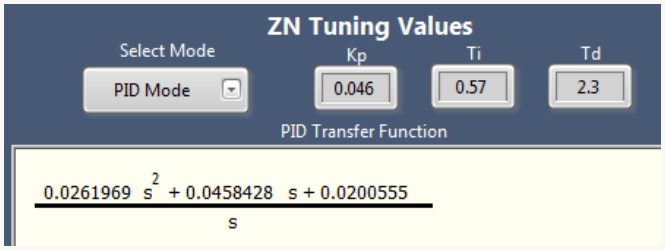
6. Stored graph will be reloaded once the experiment is selected in the ‘Schematic’ tab.



7. While cursor is moved, the FOPDT model will automatically be calculated and displayed as shown below.

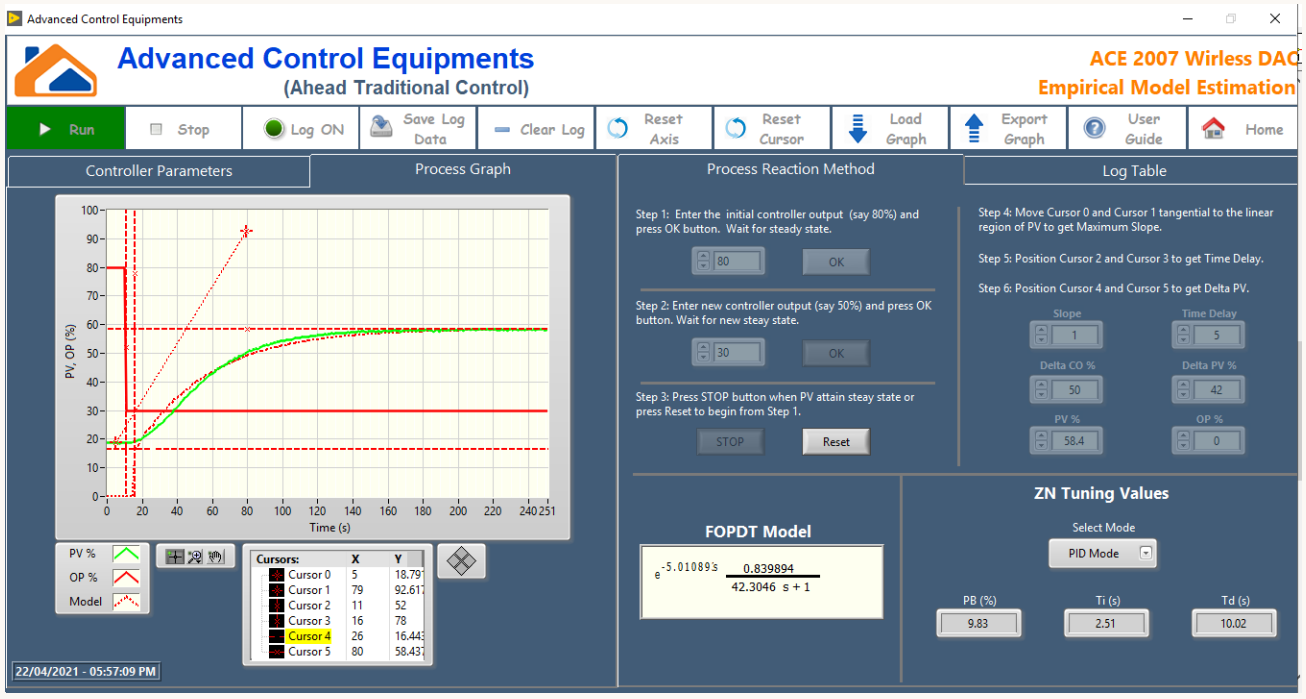


8. Similarly, the Ziegler-Nicholas tuned P, PI, PD and PID values and also the corresponding transfer function will also be displayed as shown below.



9. Select the appropriate mode and observe the text box.
10. Log On and Clear log will be used to store the data while performing the experiment.
11. Once experiment is done, go to ‘Process Schematic’ and press back.
12. A popup message will confirm the exit and other experiment can be selected and performed.
13. Note: One set of data stored in .gd format is sufficient to understand the other methods.
14. Load graph and perform other experiments.

Screenshot:



5. Component Specifications

The description of the component used in this trainer are listed below:

5.1 WIKA – Pressure Transmitter



Product	Level (Pressure) Transmitter
Manufacturer	Wika
Model	SL-1-A-MAG-ND-ZA4Z-ZZZ
Range	0-50 mbar
Input	0 to 250 mmwc
Output	4 - 12 mA (calibrated to 0 - 250 mm Head)
Power	24V DC
Transmission	2 wire

© Siemens 2020

Pressure Measurement
Pressure transmitters
for applications with advanced requirements (Advanced)
SITRANS P DS III

Technical description

1

Pressure transmitter for gauge pressure

Measured variable: Gauge pressure of aggressive and non-aggressive gases, vapors and liquids.

Measuring span (infinitely adjustable)
for DS III with HART: 0.01 bar to 700 bar (0.15 psi to 10153 psi)

Nominal measuring range
for DS III with PROFIBUS PA and FOUNDATION Fieldbus:
1 bar to 700 bar (14.5 psi to 10153 psi)

Pressure transmitters for absolute pressure

Measured variable: Absolute pressure of aggressive and non-aggressive gases, vapors and liquids.

Measuring span (infinitely adjustable)
for DS III with HART: 8.3 mbar a ... 100 bar a (0.12 ... 1450 psi a)

Nominal measuring range
for DS III with PROFIBUS PA and FOUNDATION Fieldbus:
250 mbar a ... 100 bar a (3.6 ... 1450 psi a)

There are two series:

- Gauge pressure series
- Differential pressure series

Pressure transmitters for differential pressure and flow

Measured variables:

- Differential pressure
- Small positive or negative pressure
- Flow $q \sim \sqrt{\Delta p}$ (together with a primary differential pressure device (see Chapter "Flow Meters"))

Measuring span (infinitely adjustable)
for DS III with HART: 1 mbar ... 30 bar (0.0145 ... 435 psi)

Nominal measuring range
for DS III with PROFIBUS PA and FOUNDATION Fieldbus:
20 mbar ... 30 bar (0.29 ... 435 psi)

Pressure transmitters for level

Measured variable: Level of aggressive and non-aggressive liquids in open and closed vessels.

Measuring span (infinitely adjustable)
for DS III with HART: 25 mbar ... 5 bar (0.363 ... 72.5 psi)

Nominal measuring range
for DS III with PROFIBUS PA and FOUNDATION Fieldbus:
250 mbar ... 5 bar (3.63 ... 72.5 psi)

Nominal diameter of the mounting flange

- DN 80 or DN 100
- 3 inch or 4 inch

In the case of level measurements in open containers, the low-pressure connection of the measuring cell remains open (measurement "compared to atmospheric").

In the case of measurements in closed containers, the lower-pressure connection has to be connected to the container in order to compensate the static pressure.

The wetted parts are made from a variety of materials, depending on the degree of corrosion resistance required.

Design



Front view

The transmitter consists of various components depending on the order. The possible versions are listed in the ordering information. The components described below are the same for all transmitters.

The rating plate (7, Figure "Front view") with the Article No. is located on the side of the enclosure. The specified number together with the ordering information provide details on the optional design details and on the possible measuring range (physical properties of built-in sensor element).

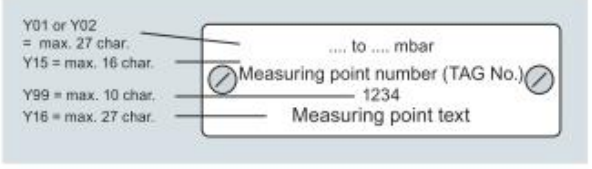
The approval label is located on the opposite side.

The enclosure is made of die-cast aluminium or stainless steel precision casting. A round cover (6) is screwed on at the front and rear of the enclosure. The front cover can be fitted with a viewing pane so that the measured values can be read directly on the display. The inlet (8) for the electrical connection is located either on the left or right side. The unused opening on the opposite side is sealed by a blanking plug. The protective earth connection is located on the rear of the enclosure.

The electrical connections for the power supply and screen are accessible by unscrewing the rear cover. The bottom part of the enclosure contains the measuring cell with process connection (5). The measuring cell is prevented from rotating by a locking screw (4). As the result of this modular design, the measuring cell and the electronics can be replaced separately from each other. The set parameter data are retained.

At the top of the enclosure is a plastic cover (1), which hides the input keys.

Example for an attached measuring point label



https://cache.industry.siemens.com/dl/files/046/109765046/att_1041177/v1/sitransp_ds3_fi01_en.pdf

5.2 Delta – Digital Temperature Indicating Controller (DTB9696CR)



Product	Temperature Controller
Product code	DTB9696CR
Manufacturer	Delta
Input Voltage	100 to 240VAC, 50/60Hz
Operation Voltage Range	85% to 110% of rated voltage
Power Consumption	5VA max.
Memory Protection	EEPROM 4K bit - non-volatile memory (number of writes: 100,000)
Display Method	2 line x 4 character 7-segment LED display Process value (PV): Red color, Set point (SV): Green color
Thermocouple:	K, J, T, E, N, R, S, B, L, U, TXK
3-wire Platinum RTD:	Pt100, JPt100 Sensor Type
Analog input:	0 ~ 5V, 0 ~ 10V, 0 ~ 20 m A, 4 ~ 20 m A, 0 ~ 50mV
Control Mode	PID, ON/OFF, Manual or PID program control (Ramp/Soak control)
Control Output	Relay output: SPDT (SPST: 1/16 DIN and 1/32 DIN size), Max. load 250VAC, 5A resistive load
	Voltage pulse output: DC 14V, Max. output current 40mA
	DC 4 ~ 20m A output (Load resistance: Max. 600)
	Linear voltage output: 0 ~ 10V
Display Accuracy	0 or 1 digit to the right of the decimal point (selectable)
Sampling Rate	Analog input: 150 msec/ per scan Thermocouple or Platinum RTD: 400 msec/per scan
RS-485 Communication	MODBUS ASCII / RTU communication protocol
Vibration Resistance	10 to 55Hz, 10m/s^2 for 10min, each in X, Y and Z directions
Shock Resistance	Max. 300m/ s^2, 3 times in each 3 axes, 6 directions
Ambient Temperature	0°C to +50°C
Storage Temperature	-20°C to +65°C
Altitude	2000m or less
Relative Humidity	35% to 80% (non-condensing)

5.3 Control Air – I/P Converter



Input	0-10 Vdc, 0-5 Vdc, 1-9 Vdc, 4-20 mA
Output Range	1-17 psig (0.07-1.2 bar), 2-60 psig (0.14-4 bar), 3-120 psig (0.2-8 bar), 3-15 psig (0.20-1 bar), 3-27 psig (0.2-1.8 bar), 3-9 psig (0.2-0.6 bar), 6-30 psig (0.40-2.0 bar), 9-15 psig (0.6-1 bar)
Max. Supply Pressure	150 psig (10 bar)
Supply Pressure Sensitivity - Low Output Range	<+ 0.1% of span per psig (<+ 0.15% of span per 0.1 bar)
Supply Pressure Sensitivity - High Output Range	<+ 0.04% of span per 1.0 psig (0.07 bar)
Flow Rate	20 scfm (566 Nl/min)
Maximum Air Consumption	0.07 scfm (1.98 Nl/min)
Repeatability	< 0.5% of span < 0.5% of span
Hysteresis	< 1.0% of span < 0.5% of span
Response Time	Dependent on pressure range – typically less than 0.25 sec. for 3-15 psig units
Media	Oil free, clean dry air filtered to 40 micron
Electrical Connection	Conduit 1/2" NPT, DIN 43650
Enclosure	Nema 1, Nema 4X (IP65)
Ambient Temperature Limits	-20 to +140 F ('-30 to 60 C)
Port Sizes	1/4 NPT (Pneumatic), 1/2 NPT (Electric)
Weight	2.1 lbs (0.94 kg.)
URL	https://www.controlair.com/product/type-500-electro-pneumatic-i-p-transducer-i-p-e-p/

5.4 Pneucon – Control Valve

PNEUCON

LOW FLOW CONTROL VALVE
(SERIES – 119)

INTRODUCTION

Low flow control valve is designed for modulating control of liquid and gases in general duty industrial application where performance, quality and small physical size are important. The basic design offers features and performance levels normally found in more expensive, larger valves.

SPECIFICATIONS

DESIGN	: ASME B16-34
SIZE	: 6 to 25 mm (1/4" to 1")
RATING	: ANSI 150.
END CONNECTION	: Screwed - BSP / NPT or Socketweld.
BODY MATERIALS	: CF8 & CF8M For Valve Size - 1/4" & 1/2"
	A216 Gr WCB, CF8 & CF8M For Valve Size - 3/4" & 1"
PLUG, SEAT RING	: S.S.316
GLAND PACKING	: PTFE.
BONNET	: CF8 & CF8M For Valve Size - 1/4" & 1/2"
	A216 Gr WCB, CF8 & CF8M For Valve Size - 3/4" & 1"
TRIM DESIGNS	: Contoured, Micro Splined.
CHARACTERISTICS	: Equal %, Linear and On-Off.
SEAT LEAKAGE	: Class IV, V & VI
	As per FCI-70.02 (ANSI B 16.104)
ACTUATOR TYPE	: Diaphragm.
ACTUATOR ACTION	: Direct / Reverse Acting.
DIAPHRAGM	: Nitrile.
SPRING RANGE	: 3 – 15 PSIG (0.2 – 1.0 Kg/cm²)
AIR SUPPLY	: 20 – 35 PSIG (1.4 – 2.5 Kg/cm²)
AIR CONNECTION	: 1/ 4" NPT



CONTROL VALVE WITH REVERSE ACTUATOR



CONTROL VALVE WITH DIRECT ACTUATOR

DESIGN FEATURES

>> Solid stem connection.

>> Suitable for low flow and corrosive atmospheres.

>> Available without positioner for fail open or fail close application.

>> Corrosion resistance actuator available as an option.

>> Investment cast body.

>> Wide flow capacity range. Cv ranges from 0.13 to 10.

PVCT-119

BUILT IN RELIABILITY

Product	Control Valve
Manufacturer	Penucon
Model	PVCT-119
Input	3 - 15 psig
Valve Size	15 mm (1/2”)
Type	Linear, Air to close, Globe 2 way
End Connection	Screwed Female
Material	Body CCS, Trim SS
Actuator	Type: Diaphragm, Travel 14.3mm, 12 Square inch
Spring Range	0.2-1
CV	0.63
Air Connection	1/ 4” NPT
URL	https://www.pneuconvalves.com/low-flow-globe-2-way-control-valve-series-119/

5.5 Eureka – Rotameter



Loflo Meters

Series ('MGS, MGE, MGZ')

Suitable for purging or low flowrate measurement. The standard mounting is 1/4" BSP / NPT rear, although several arrangements of Flanged or Hose pipe connection are offered on request.

Ranges: Between 0.6 to 160 LPH / 0.003 to 0.7 USGPM of Water at Ambient Temperature. Between 1 to 5000 NLPH / 0.001 to 3.1 SCFM of Air at Ambient Temperature, 760 Torr.

Accuracy: ±3% of full flow. / ±5% of full flow. Standard Models are available

with built in needle valve.
Accessories: 1. DP Regulator.

2. Panel Mounting Clamps.

3. High & Low Flow Alarms. Special sterilizable Loflo meter for Auto-clave applications is also available against specific enquiries.

Accuracy	±2% of Full Scale
Repeatability	10:1 Turndown Ratio
Liquid Flow Range	10 to 100 LPH (Water)
Gas Flow Range	1 to 5000 NLPH (Air)
Body Materials	MS / SS304 / SS316 / PP / PVC
Metering Tube	Borosilicate Glass
Float Material	SS316, PTFE, or Ceramic
Gaskets/O-Rings	Neoprene, Viton, PTFE, or Silicon
End Connections	1/4" BSP / NPT (Rear or Vertical)
Max Temperature	Up to 150 C (depending on material)
Max Pressure	Up to 10 Bar (standard)
URL	https://5.imimg.com/data5/SELLER/Doc/2022/12/XG/HG/CQ/1955678/rotameter.pdf

6. Controller Settings & Modbus Address

6.1 Default Settings

- There are three modes of operation: operation, regulation and initial setting. When power is applied, controller gets into the operation mode. Press the SET key to switch to regulation mode. If the SET key is pressed for more than 3 seconds, controller will switch to the initial setting mode. Pressing the SET key while in the regulation mode or initial setting mode, forces the controller to return to the operation mode.
- PV/SV: Sets the temperature set point and displays the temperature process value. Use ▼ ▲ keys to set the temperature set point.
- Setting method: While in any function mode, press the ↵ key to select the desired function and use the ▼ ▲ keys to change settings.
- Press SET key to save the changes. The next flow chart shows how to switch for settings and internal functions:



*Note: Values highlighted in RED are not to be altered

Mode	Parameter	Default Value
Regulation	AT	oFF
	Pcd0	50.0
	Su0	50.0
	P0	20.0
	c0	30
	d0	1
	coF0	0.0
	cLPd	0.5
	tPoF	0.0
	crhc	100
	crLo	0

Operation	r-S	run
	SP	1
	Loc	oFF
	out 1	0.0

Initial Setting	cnPt	NA4
	tP-h	100.0
	tP-L	0.0
	ctrL	Pid
	S-hc	cooL
	ALA1	0
	ALA2	0
	ALA3	0
	SALA	oFF
	coSh	on
	c-SL	rtu
	c-no	1
	bPS	9600
	LEn	8
	PrtY	nonE
	StoP	1

6.2 Modbus Addresses

Sl. No.	Description	Memory Address	MODBUS Address
1	Process Variable (PV)	1000H	44097
2	Set Point (SP)	1001H	44098
3	Auto / Manual	1005H	44102
4	Action	1006H	44103
5	Proportional Band (PB)	1009H	44106
6	Integral Time (Ti)	100AH	44107
7	Derivative Time (Td)	100BH	44108
8	Bias	100CH	44109
9	Controller Output (OP)	1012H	44115
10	Auto Tuning	813H	42068

7. About Us

7.1 Introduction

With a strong focus on applied learning, Advanced Control Equipments develops miniaturized industrial working models for laboratory environments. We design and manufacture laboratory modules that provide hands-on training across disciplines including mechanical, electrical, electronics, mechatronics, instrumentation, and chemical engineering.

With over two decades of experience, ACE is committed to delivering industry-standard equipment at affordable prices. Our product range includes process control systems, instrumentation systems, mechatronics setups, PLC and microcontroller-based systems, home automation solutions, and more.

Using our powerful in-house software, we support both real-time control and model-based simulation and control. Our user-friendly simulation / hardware interface tools help aspiring engineers understand key concepts such as mathematical modelling, controller design and analysis, PID tuning, and model-based control strategies. We also enable users to implement custom control algorithms, in addition to built-in advanced control techniques such as Fuzzy Control, Gain Scheduling, and Model Predictive Control.

7.2 Vision and Values

Vision

To develop industry-standard laboratory products in the field of instrumentation and process control, enabling effective and meaningful teaching–learning experiences.

Values

- Over two decades of experience in laboratory equipment design and development
- Capability to deliver solutions for a wide range of instrumentation and process control applications
- User-friendly hardware - software interfaces
- Customizable product designs based on customer requirements
- Commitment to timely delivery as promised

7.3 Product Range

From fundamental concepts to advanced model-based control, Advanced Control Equipments offers a wide range of equipment and tools. With strong expertise in both hardware and software, ACE supports faculty, students, and the research community in areas such as process control, modelling and simulation, instrumentation systems, chemical and mechanical systems, data acquisition, and the Internet of Things. Based on the application, the products are grouped as given below:

Product Series	Description	
ACE 100 Series	Basic Process Trainers	
	ACE 101	Level Process Trainer
	ACE 102	Flow Process Trainer
	ACE 103	Pressure Process Trainer
	ACE 104	Temperature Process Trainer
	ACE 105	Multi Process Trainer (Level, Flow, Pressure & Temperature)
	ACE 106	Cascade Process Trainer (Level + Flow)
	ACE 107	Cascade Process Trainer (Temperature + Flow)
	ACE 108	Multi Variable Process Trainer (Level, Flow, Cascade, Feedforward, Ratio)
	ACE 109	Coupled Tank Interacting Level Process Trainer
	ACE 110	Four Tank Level Process Trainer
	ACE 111	Five Tank Level Process Trainer
ACE 200 Series	Advanced Process Trainers	
	ACE 201	Interdependent Variable MIMO Process Trainer
	ACE 202	SCADA, DDC and HMI Trainer
	ACE 203	Advanced Control Strategies Trainer
	ACE 204	MIMO Process Control Trainer
	ACE 205	Industrial Communication Protocol Trainer
	ACE 206	Interacting Quadruple System Trainer
	ACE 207	Hybrid Tank System
	ACE 208	Universal Process Trainer
	ACE 209	Universal Process Measurement & Control Trainer
	ACE 211	Distillation Column Trainer
	ACE 212	Continuous Stirred Tank Reactor Trainer
	ACE 213	Universal Process Trainer
	ACE 214	Distributed Control System Trainer
	ACE 215	Liquid-Liquid Extraction Trainer
	ACE 216	Packed Bed Column Trainer
	ACE 217	Computerized Bioreactor System
ACE 300 Series	Non-Linear Process Control Trainers	
	ACE 301	pH Process Trainer
	ACE 302	Conical Tank Level Process Trainer
	ACE 303	Spherical Tank Level Process Trainer
	ACE 304	Conical & Spherical Tank Level Process Trainer
	ACE 305	Conical (Spherical) Two Tank MIMO Process Trainer
	ACE 306	Linear & Nonlinear Multi Tank Level Process Trainer (5 Cylindrical + 1 Spherical + 1 Conical)
	ACE 307	Linear & Nonlinear Multi Shape Level Process Trainer

ACE 400 Series	Process Simulators	
	ACE 401	Two Position & PID Simulator
	ACE 402	Model based Closed loop simulator with PID controller
	ACE 403	Empirical Model Estimator
	ACE 404	Model Analysis & PID Controller Design
	ACE 405	PID Tuning Techniques
	ACE 406	Cascade Control Simulator
	ACE 407	Feedforward + Feedback Simulator
	ACE 408	Ratio Control Simulator
	ACE 409	Non-Interacting Level Tank Control Simulator
	ACE 410	Fuzzy Control Design & Simulator
	ACE 411	Non-Interacting Level Tank Simulator with Smith Predictor
	ACE 412	Internal Model Control (IMC) Simulator
ACE 500 Series	Instrumentation & Measurement Trainers	
	ACE 501	I/P and P/I Converter
	ACE 502	Control Valve Characteristics Trainer
	ACE 503	Temperature Measurement Trainer
	ACE 504	Flow Measurement Trainer
	ACE 505	Level Measurement Trainer
	ACE 506	Pressure Measurement System
	ACE 507	First Order & Second Order Systems
	ACE 508	Bourdon Tube Pressure Gauge Tester
	ACE 509	Interacting and Non-Interacting Trainer
	ACE 510	Flapper Nozzle System
	ACE 511	Multifunction Calibrator
	ACE 512	Speed / Torque Measurement Trainer
	ACE 513	Humidity Measurement and Control Setup
	ACE 514	Electronic PID Controller Setup
	ACE 515	Pressure, Flow & Level measurement trainer
	ACE 516	Solid Flow Measurement and Control Trainer
	ACE 517	Conductivity Measurement Trainer
ACE 600 Series	Mechanical Systems	
	ACE 601	Lagged pipe assembly
	ACE 602	Heat Exchangers (Computerised / Non-Computerised)
	ACE 603	Transesterification Reactor
	ACE 604	Two Tank Heater Setup
	ACE 605	Dryer Setup
ACE 700 Series	IoT Systems	
	ACE 701	Remote Monitoring Station

	ACE 702	Compact Universal IoT Trainer Kit
	ACE 703	IoT Trainer Kit with Raspberry Pi & Arduino
	ACE 704	IoT Trainer Kit with Raspberry Pi & ESP32
	ACE 705	IoT Trainer Kit with IoT2040 Gateway
ACE 800 Series	PLC Based Systems	
	ACE 801	Compact PLC Trainer
	ACE 802	Bottle Filling Station
	ACE 803	Mixing Process Station
	ACE 804	Temperature Process Station
	ACE 805	Material Sorting Station
	ACE 806	Motor Control Station
	ACE 807	Elevator Control Station
	ACE 808	Pick & Place Robot Station
	ACE 809	Traffic Light Control Station
	ACE 810	Flow Process Trainer
	ACE 811	Level Process Trainer
	ACE 812	Temperature Process Trainer
	ACE 813	Pressure Process Trainer
	ACE 814	Pneumatic Control Trainer Using PLC
	ACE 815	Annunciation System Using PLC
ACE 2000 Series	Data Acquisition Systems	
	ACE 2001	Voltage Module 8AI / 2AO
	ACE 2002	Current Module 4AI / 2AO
	ACE 2003	Dual Loop PID Controller (Voltage Module)
	ACE 2004	Dual Loop PID Controller (Current Module)
	ACE 2005	Voltage (8AI / 2AO) / Current (4AI / 2AO) Module
	ACE 2006	Wireless Voltage Module 6AI / 2 DI / 2AO
	ACE 2007	Wireless Current Module 4AI / 2AO
	ACE 2008	Wireless Dual Loop PID Controller (Voltage Module)
	ACE 2009	Wireless Dual Loop PID Controller (Current Module)
	ACE 2101	Edge IoT DAQ Module
	ACE 2102	Industrial Gateway DAQ Module

7.4 Customer List

From fundamental concepts to advanced model-based control, Advanced Control Equipments offers a wide range of equipment and tools. With strong expertise in both hardware and software, ACE supports faculty, students, and the research community in areas such as process control, modelling and simulation, instrumentation systems, chemical

No.	State	Place	Institute
1	Andhra Pradesh	Kakinada	University College of Engineering, JNTU
2	Andhra Pradesh	Srikakulam	GMR Institute of Technology
3	Andhra Pradesh	Vijayawada	PVP Siddhartha Institute of Technology
4	Andhra Pradesh	Visakhapatnam	Andhra University College of Engineering
5	Assam	Guwahati	Indian Institute of Technology
6	Assam	Kokrajhar	Central Institute of Technology
7	Goa	Goa	Birla Institute of Technology, Goa Campus
8	Goa	Goa	Padre Conceicao College of Engineering
9	Goa	Verna	Agnel Institute of Technology and Design
10	Jammu Kashmir	Jammu	Indian Institute of Technology
11	Jharkhand	Ranchi	Birla Institute of Technology, Mesra
12	Karnataka	Bangalore	BMS College of Engineering
13	Karnataka	Bangalore	Dayananda Sagar College of Engineering
14	Karnataka	Bangalore	R V College of Engineering
15	Karnataka	Belagavi	KLE Dr. M. S. Sheshgiri College of Engineering and Technology
16	Karnataka	Dharwad	Sri Dharmasthala Manjunatheshwara College of Engineering & Technology (SDM, Dharwad)
17	Karnataka	Mangalore	Mangalore Institute of Technology and Engineering
18	Karnataka	Manipal	Manipal Institute of Technology
19	Karnataka	Tumkur	Siddaganga Institute of Technology
20	Kerala	Calicut	National Institute of Technology
21	Maharashtra	Ahmednagar	Dr. Vithalrao Vikhe Patil College of Engineering
22	Maharashtra	Ahmednagar	Pravara Rural Engineering College
23	Maharashtra	Mumbai	Bharatiya Vidya Bhavan’s Sardar Patel Institute of Technology
24	Maharashtra	Nanded	Sh. Guru Gobind Singhji College of Engineering & Technology
25	Maharashtra	Nasik	NDMVPS's KBT College of Engineering
26	Maharashtra	Navi Mumbai	Bharati Vidyapeeth's College of Engineering

27	Maharashtra	Pune	Dr. D. Y. Patil Institute of Technology
28	Maharashtra	Pune	Vishwakarma Institute of Technology
29	Odisha	Rourkela	National Institute of Technology
30	Punjab	Ropar	Indian Institute of Technology
31	Rajasthan	Jaipur	Malaviya National Institute of Technology Jaipur (MNIT)
32	Rajasthan	Pilani	Birla Institute of Technology, Pilani
33	Tamil Nadu	Chennai	Apex Service and Instruments
34	Tamil Nadu	Coimbatore	Amrita Vishwa Vidyapeetham
35	Tamil Nadu	Coimbatore	PSG College of Technology
36	Tamil Nadu	Coimbatore	PSG Institute of Technology and Applied Research (PSG iTech)
37	Tamil Nadu	Mannampandal (Mayiladuthurai)	AVC College of Engineering
38	Tamil Nadu	Perundurai	Kongu Engineering College
39	Tamil Nadu	Tiruchirappalli	National Institute of Technology
40	Tamil Nadu	Tiruchirappalli	Saranathan College of Engineering
41	Tamil Nadu	Vellore	Vellore Institute of Technology
42	Telangana	Medak	Medak College of Engineering and Technology
43	Telangana	Warangal	National Institute of Technology
44	Telangana	Hyderabad	Birla Institute of Technology & Science Pilani, Hyderabad


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istan
Islamabad
اسلام آباد
LADAKH
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Indian Institute Of Technology–Ropar (IIT–R...
PUNJAB
Pakistan
Indian Institute Of Technology Roorkee
41
Malaviya National Institute of Technology...
ARUNACHAL PRADESH
Central Institute of Technology, Kokrajhar
Indian Institute of Technology Guwahati
Indian Institute of Technology (BHU) Varana...
Karachi
کراچی
Adani University
MANIPUR
Birla Institute of Technology - Mesra
Atmiya University
India
MIZORAM
AMNS India HRC Gate
Siddaganga Institute Of Technology
Myanmar (Burma)
ODISHA
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GMR Institute of Technology
Naypyida
Yangon
Agnel Institute of Technology and Design
University College of Engineering JNTUK Naras...
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NIT Calicut
A V C COLLLEGE OF ENGINEERING
Andaman Sea
Muthoot Institute of Technology & Science